

The 58th Annual Graduate Student Symposium

Photo:

Hosted by the Department of Geosciences
April 16th – 17th, 2026

58th Annual Graduate Student Symposium

Hosted by the Department of Geosciences.

The Graduate Student Symposium is a forum where students present their research or research proposal to faculty, friends, and peers. It is hosted by the Department of Geosciences and is open to graduate students involved in geosciences research. The symposium format stimulates research discussion, allows students to practice for national meetings, and helps students improve their presentation skills. It assists both the Department and Penn State in maintaining and strengthening their reputations for giving high quality talks and poster presentations at national and international meetings.

The Graduate Symposium Committee wishes to thank the students for sharing their work and the post-docs and faculty for providing constructive feedback. The Committee thanks the Department of Geosciences for hosting this Symposium.

Graduate Student Symposium Committee

Ava Spangler, Arthur Lo, Amina Abdulsalam, Liz Tofte, Doni Angelica, Martina Adjei-Yeboah

Data Visualization:

Event Schedule

Thursday 16th April

Opening Remarks – 8:50 am

Oral Session 1 – 9:00 to 10:30 am

Break – 10:30 to 10:45 am

Oral Session 2 – 10:45 am to 12:15 pm

Lunch Break – 12:15 pm to 1:00 pm

Poster Session 1 – 1:00 to 2:30 pm

Break – 2:30 to 3:00 pm

Oral Session 3 – 3:00 to 4:00 pm

Friday 17th April

Poster Session 2 – 9:00 to 10:30 pm

Break – 10:30 to 11:00 am

Oral Session 4 – 11:00 am to 12:00 pm

Lunch Break – 12:00 pm to 1:00 pm

Oral Session 5 – 1:00 to 2:15 pm

Break – 2:15 to 2:45 pm

Oral Session 6 – 2:45 to 4:00 pm

The Peter Deines Lectureship

The first-place award for an oral presentation by a post-comprehensive Ph.D. student is designated the Peter Deines Lectureship for the following academic year.

This award was started in 2004 to represent the tremendous amount of respect and admiration the graduate students in the Department of Geosciences had for Dr. Peter Deines, who that year was stepping down from the position of Graduate Program Chairman. Recipients of the honor are invited to give a departmental colloquium talk during the proceeding academic year.

The department and the world lost a great man and wonderful person when Peter passed away on February 2, 2009. It is with great pride that the Graduate Student Symposium continues the tradition born in 2004.

Recipients:

2025-26: Kayla Irizarry
2024-25: Safiya Alpheus
2023-24: Samuel Shaheen
2022-23: Shelby Bowden
2021-22: Julia Carr
2020-21: Graduate Student Symposium Canceled due to COVID-19 pandemic
2019-20: Allison Fox
2018-19: Beth Hoagland
2017-18: Matthew Herman
2016-17: Rosie Oakes
2015-16: John Leeman
2014-15: Ashlee Dere
2013-14: Jonathon Schueth
2012-13: Elizabeth Herndon
2011-12: Bryan Kaproth
2010-11: Tim Fischer
2009-10: Aaron Diefendorf and Bryn Kimball
2008-09: Daniel Hummer
2007-08: Gavin Hayes
2006-07: Christina Lopano
2005-06: Shawn Goldman and Courtney Turich
2004-05: Margaret Benoit

The Peter Deines Lectureship



Peter Deines (4/02/36 - 2/02/09) earned a Geologen Vordiplom at the Rheinische Friedrich Wilhelms Universität, Bonn, Germany in 1959, an M.S. (1964) and a Ph.D. (1967) in Geochemistry and Mineralogy from Penn State University. Since 1967, and after 2004, as an Emeritus Professor, he was a member of the Geological Science Faculty of the Pennsylvania State University. He earned an international reputation for his geochemical research, teaching, and science administration. Recognition came in teaching awards, election to the University Senate, in which he served for 24 years, and election especially to Treasurer of the International Geochemical Society. In that office, he was so effective that he was awarded a unique Honorary Life Membership for his financial management of the society. He was a principal organizer of that Society's primary international meetings, the famous Goldschmidt Conferences.

With his gift for organization, he also served the Department of Geosciences on most of its committees. He served as its Graduate Program Chairman, while also administering committees for the College of Earth and Mineral Sciences, primarily for scholarships. Most important was his commitment to the University Academic Senate, in which he served in 28 committee posts, including its Chair for 1990-91; and to the University, on 34 committees and commissions, including University Ombudsman since 2006. He also was elected President of the Faculty-Staff Club. Dr. Deines' research centered on precise explanations of natural variations in stable isotope abundances as means of understanding geologic processes. Results were presented in lectures throughout the world and in over 60 published papers. His illustrated book, "Solved Problems in Geochemistry", was polished by his teaching of eight graduate courses and is available on the web especially for graduate students.

A 40-year member of the Nittany Valley Symphony, Peter will be missed for his finesse with violin and viola.

Table of Contents

Thursday 16th April	6-29
Session 1: Oral Session 1	6
Session 2: Oral Session 2	12
Session 3: Poster Session 1	17
Session 4: Oral Session 3	25
Friday 17th April	30-49
Session 1: Poster Session 2	30
Session 2: Oral Session 4	41
Session 3: Oral Session 5	46
Session 4: Oral Session 6	46

Thursday 16th April

Oral Session 1

9:00 am – 10:30 am

9:00 am – 9:15 am

Cris Kocian

M.S. student, 2nd year

Experimental Constraints on the Origin of ¹³C-Depleted Martian Sediments in Gale Crater

9:15 am – 9:30 am

Liz Tofte

M.S. student, 2nd year

Quantifying Seasonal Sea Ice Melt and Sediment Plume Development at Arctic River Mouths

9:30 am – 9:45 am

Renan Beckman

Ph.D. student, 2nd year, Pre-Comps

Exploring Zircon Dark Zones: New Insights from the Ruby Star Granodiorite

9:45 am – 10:00 am

Lillian Miller

Ph.D. student, 2nd year, Pre-Comps

Using InSAR, SAR Offset Tracking, and GNSS to Investigate the August 28, 2020 Bedrock Landslide Within the Irazú-Turrialba Volcanic Complex, Costa Rica

10:00 am – 10:15 am

Ethan Heidtman

Ph.D. student, 2nd year, Pre-Comps

Modeling and mitigation of riverine saltwater intrusion hazards using reservoir operations: a case study in the tidal Susquehanna River

10:15 am – 10:30 am

Arthur Lo

Ph.D. student, 2nd year, Pre-Comps

Analysis of Ground Deformation Preceding the December 2020 – May 2021 Summit Eruption at Kīlauea Volcano, Hawai'i

Experimental Constraints on the Origin of ^{13}C -Depleted Martian Sediments in Gale Crater

Cris Kocian¹, Christopher House¹, Jennifer Stern², Steven M. Bates²

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

²*NASA Goddard Space Flight Center, Greenbelt, MD, USA*

In Gale Crater, the Curiosity rover found organic molecules with a wide range of carbon isotopic compositions. Methane (CH_4) released during pyrolysis by the Sample Analysis at Mars (SAM) instrument exhibits $\delta^{13}\text{C}$ values in the range of $-137 \pm 8\text{‰}$ to $+22 \pm 10\text{‰}$. Ten of the twenty-four samples in this dataset are strongly depleted in ^{13}C ($\delta^{13}\text{C} < -70\text{‰}$). Two hypotheses to explain this ^{13}C depletion with abiotic processes are 1) contamination of the derivatizing agent N-tert-butyldimethylsilyl-N-methyltrifluoroacetamide (MTBSTFA) and 2) photochemical deposition of formaldehyde (HCHO). MTBSTFA, which is present in the background of the SAM instrument, has a $\delta^{13}\text{C}$ value of about -35‰ and would produce CH_4 if pyrolyzed. Alternatively, ultraviolet radiation in the Martian atmosphere photolyzes carbon dioxide (CO_2) into carbon monoxide (CO) depleted in ^{13}C with a $\delta^{13}\text{C}$ of approximately -150‰ . If this ^{13}C -depleted CO is further photochemically reduced to HCHO it could deposit onto the surface and be detected by Curiosity.

To test these hypotheses, we simulated SAM pyrolysis conditions in the laboratory and measured CH_4 production and carbon isotope fractionation from candidate compounds. Tetrakis(trimethylsilyl)silane (TKS) was used as an analog for MTBSTFA. Pyrolysis of TKS at low (75°C) and high (455°C) temperature cuts produced CH_4 with $\delta^{13}\text{C}$ values from -49‰ to -43‰ , which are not depleted enough to account for lower SAM measurements. In contrast, pyrolysis of HCHO at the same temperature cuts produced primarily CO and minimal CH_4 . Despite lower CH_4 yields, HCHO derived from ^{13}C -depleted atmospheric CO could reproduce the SAM $\delta^{13}\text{C}$ values if the isotopic composition of photolyzed CO is retained. These results support photochemically produced HCHO as a more likely abiotic source of the ^{13}C -depleted CH_4 detected by Curiosity than MTBSTFA contamination.

Quantifying Seasonal Sea Ice Melt and Sediment Plume Development at Arctic River Mouths

Liz Tofte¹, Anastasia Piliouras¹

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

The Arctic is warming at more than four times the global average. This warming has led to significant changes in Arctic river hydrology, including increased discharge, greater sediment supply, and earlier spring freshet onset. These upstream changes affect the coastal zone – dynamic land–ocean interfaces where seasonal riverine and oceanic processes regulate primary productivity – making them particularly vulnerable to climate change. Coastal sea ice melt and sediment plume development are two seasonal processes that directly respond to freshwater, heat, and sediment export from rivers, yet they remain under-observed. Quantifying how these processes behave and change over time can provide insight into how the coastal zone is responding to changing hydrology.

This study investigates six major Arctic river mouths: the Yukon, Colville, and Mackenzie in North America, and the Ob-Yenisei, Kolyma, and Lena in Eurasia. The objectives are to (1) characterize typical seasonal patterns of coastal sea ice melt and sediment plume development, (2) assess changes over two decades, and (3) compare variability between systems. Sea ice and plume behaviors were analyzed using MODIS-derived sea ice products and MODIS imagery, respectively.

Results show that river mouths follow two primary ice melt regimes: initiating directly at the shoreline *or* at offshore flaw leads, suggesting differences in dominant processes driving coastal ice melt initiation. As temperatures continue to increase into the summer, sediment plumes develop, with turbidity peaking in early summer following snowmelt and occasionally in late summer due to storm-driven resuspension. Eurasian rivers show significant trends in sea ice behavior, including a shortening melt season at the Lena and Ob–Yenisei and earlier landfast melt at the Kolyma. Across all systems, sediment delivery to the coast is occurring earlier, with significantly earlier plume onset at the Lena, Kolyma, and Mackenzie. Overall, Arctic coastal processes are shifting in response to rapid warming, though the magnitude and timing of change vary among river systems, reflecting strong local controls.

Exploring zircon dark zones through cathodoluminescence and trace elements: New insights from the Ruby Star Granodiorite, Arizona

Renan Beckman¹, Herve Rezeau², Erik Schoonover¹, Maureen Feineman¹, Jesse Reimink¹

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

²*Earth Resources Laboratory, Department of Earth, Atmospheric and Planetary Sciences, Massachusetts Institute of Technology, Cambridge, MA, USA*

Copper is critical to the global energy transition, and demand is rising rapidly. The main global copper sources are Porphyry Copper Deposits (PCDs). New PCDs are becoming more challenging to find, and major discoveries from the past decade are being found deeper in the subsurface. To improve exploration strategies, a holistic understanding of the processes that drive PCD emplacement is required.

Recent work has focused on accessory igneous minerals (e.g. zircon, apatite) within intrusions associated with PCDs as recorders of magmatic and hydrothermal evolution. Zircon textural features revealed by Cathodoluminescence (CL) record variations in the chemical evolution through its characteristic oscillatory growth zoning. This study specifically focuses on zircon “dark” growth zones that are commonly observed in porphyry intrusions, yet poorly studied (e.g., Bajo de la Alumbrera; Buret et al., 2016). In Bajo de la Alumbrera, zircon dark zones display anomalously high concentrations of uranium and rare earth elements (REE), and are interpreted as relics of a mafic recharge event (Buret et al., 2016).

To test petrogenetic models for zircon dark zone growth, and their possible usefulness as a geochemical vector for PCDs, zircon grains from the Ruby Star Granodiorite associated with world-class sierrita PCD were depth profiled via Laser Ablation Split Stream Inductively Coupled Mass Spectrometer (LASS-ICP-MS). Zircons from the Ruby Star Granodiorite commonly exhibit dark growth zones characterized by enriched U, Th, and REEs. After rotating these grains 90° and polishing to expose the depth profile, we use SEM imaging and electron microprobe analyses to evaluate the morphology and chemistry of zircon grains. Here we present these findings and discuss their implications for the petrogenetic mechanisms that form zircon dark zones in intrusions associated with PCDs, and evaluate their potential to better understand emplacement and as a geochemical indicator to identify “fertile” intrusions.

Using InSAR, SAR Offset Tracking, and GNSS to Investigate the August 28, 2020 Bedrock Landslide Within the Irazú-Turrialba Volcanic Complex, Costa Rica

Lillian Miller¹, Dr. Christelle Wauthier¹, Young Cheol Kim¹, Cyril Muller², Esteban Chaves²

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

²*Observatorio Vulcanológico y Sismológico de Costa Rica (OVSICORI-UNA), Universidad Nacional, Costa Rica*

The presence of landslides in volcanic settings is a significant threat, as large landslides on volcanic edifices can lead to cascading hazards, including smaller landslides, debris avalanches, or magma chamber depressurization. On August 28, 2020, the Volcanological and Seismological Observatory of Costa Rica (OVSICORI) observed a landslide in a valley adjacent to Irazú Volcano. The landslide occurred within the Irazú-Turrialba Volcanic Complex outside of San José, Costa Rica. Turrialba, the second volcano within the complex and just 10 km from Irazú, saw renewed activity beginning in 2010 and continuing through 2022 and has sparked concern of potential reactivation of Irazú and associated hazards. The cause of this landslide is currently unknown, but the renewed activity at Turrialba hints that the landslide may have been driven by underground magmatic activity within the volcanic complex. We propose to leverage optical and radar imagery in tandem with Global Navigation Satellite System (GNSS) positional data to answer: **(1)** was the August 28, 2020 landslide initiated by renewed magmatic movement below the Irazú-Turrialba Volcanic Complex? And **(2)** was subsidence a precursor to the landslide? **We hypothesize that the landslide was not caused by renewed magmatic movement and that subsidence was a precursor to the landslide.** GNSS results show the landmass experienced 3.5 m of accelerating subsidence over three years leading up to the landslide, with 4 m of subsidence occurring during the landslide. However, Interferometric Synthetic Aperture Radar (InSAR) analysis of Sentinel-1 C-band SAR data reveals > 1 m of subsidence across the same period, likely due to high rates of deformation, fast temporal decorrelation, and vegetation within the volcanic complex. Continued work is being done to leverage offset tracking with Planet optical data and high-resolution COSMO-SkyMed X-band SAR data to capture the displacement and more accurately match the GNSS displacement of the landmass.

Modeling and mitigation of riverine saltwater intrusion hazards using reservoir operations: a case study in the tidal Susquehanna River

Ethan Heidtman¹ and Antonia Hadjimichael^{1,2}

¹*Penn State University Department of Geosciences*

²*Penn State University Earth and Environmental Systems Institute*

Multi-purpose reservoirs are critical infrastructure for the management of water resources, and reservoir operations must often balance multiple competing water demands. Climate change is an important threat to the resiliency of water resources in these reservoirs, as it can limit the efficacy of historical operations strategies as well as water demands and availability. The Conowingo Reservoir, a large multi-purpose reservoir located in Pennsylvania and Maryland along the Susquehanna River, has been managed actively for decades, yet current operating policies fail to adequately account for internal system variability and emerging threats to regional water resource reliability. Havre de Grace, Maryland, a small community at the mouth of the Susquehanna River, sources drinking water from the river, and during periods of low riverine freshwater flow and certain weather conditions, saline waters from Chesapeake Bay can intrude upstream, degrading public water supply and corroding infrastructure.

To address this emerging problem, we first develop a statistical model for saltwater intrusion at the Havre de Grace drinking water intake using historical discharge, tide, and wind data from 2007 – 2024. Salinity responses to wind and tidal forcing under low flow conditions are rare and nonlinear, such as in October 2016, when drinking water exceeded the EPA Secondary Standard for Total Dissolved Solids for a 2-week period. These events highlight the need for targeted reservoir releases prior to and during salinity events, and results indicate that the model can identify intrusion-prone conditions more than 10 days in advance. Future work will incorporate this forecast model into a simulation-optimization framework for the Conowingo, training adaptive, state-aware, and dynamic release policies that release water from the reservoir downstream to flush out intruding saltwater while continuing to satisfy pre-existing regional water demands.

Analysis of Ground Deformation Preceding the December 2020 – May 2021 Summit Eruption at Kīlauea Volcano, Hawai'i

Arthur Wan Ki Lo¹, Christelle Wauthier^{1,2}, Young Cheol Kim¹

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

²*Institute for Computational and Data Sciences, The Pennsylvania State University, University Park, PA, USA*

The 2020 summit eruption at Kīlauea Volcano, Hawai'i marked a new phase of eruptive activity at Kīlauea's summit following the end of the 2018 Lower East Rift Zone eruption. Seismicity and ground deformation had been reported prior to the eruption, including an intrusion beneath the summit caldera on 2 December 2020. To investigate the geometry and volume of the 2 December intrusion, we processed interferograms using ascending and descending orbit Sentinel-1 SAR data with acquisition dates spanning 2 December 2020. To investigate long-term summit deformation prior to the eruption, we also processed Interferometric Synthetic Aperture Radar (InSAR) time-series of Kīlauea using data from Sentinel-1 satellites (January to mid-December 2020).

For the interferograms, we measured line-of-sight (LOS) displacements ranging from about 14 to -2.8 cm and 7 to -4.2 cm respectively for the ascending and descending orbit interferogram. We inverted the LOS displacements and modelled the intrusion using analytic solutions for point source, rectangular dislocation, penny-shaped crack and spheroid models in an elastic half space. The deformation coincides with the location of a linear feature within the summit caldera, which suggests that deformation observed in early December 2020 could be associated with movement along a caldera fault.

Using the time-series, we found a steady increase in LOS displacement within the caldera. We separated the time-series into 3 periods: P1 (January – early November), P2 (early November – early December), and P3 (early December – mid-December). We observed distinct deformation patterns, including localized deformation within Halema'uma'u Crater during P1, a marked increase in displacement rates within the caldera for P2 and deformation likely associated with the intrusion in P3. The pattern of the localized deformation during P1 coincides with the location of the eruptive vents which formed during the December 2020 eruption. This may represent a pathway along which magma was transported for the eruption.

Thursday 16th April

Oral Session 2

10:45 am – 12:15 pm

10:45 am – 11:00 am

Dan Cove

Ph.D. student, 2nd year, Pre-Comps

Laboratory methods to constrain the habitability of extreme environments

11:00 am – 11:15 am

Mingxi Hu

Ph.D. student, 2nd year, Pre-Comps

Paleohydraulics of cyclonic storm deposits suggest that the equatorial climate of Earth in the Pennsylvanian was not cold

11:15 am – 11:30 am

Ava Spangler

Ph.D. student, 3rd year, Pre-Comps

Collaborative Development of a Pluvial Flood Model and Adaptation Strategies to Advance Resilience in Baltimore

11:30 am – 11:45 am

Bailey Hoplight

Ph.D. student, 2nd year, Pre-Comps

Elevated CO2 Effects on Water Isotope Fractionation in Trees

11:45 am – 12:00 pm

Mila Matney

Ph.D. student, 2nd year, Pre-Comps

Orbitrap-MS Characterization of Polycyclic Aromatic Hydrocarbons on the Asteroid Bennu

12:00 am – 12:15 pm

Alejo Giraldo

Ph.D. student, 3rd year, Pre-Comps

Three new Atherospermataceae species from the middle Eocene add 35 million years to the family's Australian history

Laboratory methods to constrain the habitability of extreme environments

Daniel Cove¹, Chris House¹

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

Habitability, an environment's ability to support life on some scale, is a concept that is pervasive in our discussions of planets in outer space. Whether the focus is on distant exoplanets, neighboring worlds in our solar system, or the planet Earth itself at various points in the past, physicochemical conditions which constitute habitability are constantly redefined and speculated upon. One characteristic shared by each of these example locations is that they are not readily accessible *in situ*, whether due to a spatial or temporal barrier. For observed or theorized environments in such places, the development of laboratory simulations is necessary.

This study develops a series of laboratory assays that can be subjected to simulations of exotic environments to constrain habitability along a continuum from a single aqueous biological reaction to complete cell division. The assay representing the low end of the habitability spectrum is activity of the bioessential enzyme phosphatase, which cleaves phosphate ions from larger substrates to eventually create ATP for cellular energy storage. Such enzyme activity can be observed and quantified in lab experiments using ³¹P NMR spectroscopy and represents the minimum biological reaction for the host environment to be considered habitable. The higher end of the habitability spectrum is represented by the replication of a (likely extremophilic) bacterial cell. The survival and replication of a bacterial culture subject to simulated conditions can be monitored using both fluorescent dye staining techniques coupled with microscopy and labelled isotope incorporation coupled with Raman spectroscopy. Creating a reasonable lab simulation of exotic planetary conditions and using these habitability proxies to work from one end of the continuum to the other will allow for better understanding of habitability as we know it and help to clarify key astrobiological target locations across the solar system and beyond for future space missions.

The Paradox of Cyclonic Storms in a Cold Pennsylvanian Equatorial Region

Mingxi Hu¹, Paul M. Myrow², and Michael P. Lamb³

¹*Department of Geosciences, The Pennsylvania State University*

²*Department of Geology, Colorado College*

³*Division of Geological and Planetary Sciences, California Institute of Technology*

Earth has experienced extreme past climates, and the Pennsylvanian equatorial climate has been widely debated: while traditionally viewed as warm, recent studies controversially suggested widespread cold conditions and low-altitude glaciation. One tool that has not been widely evaluated is the state of the sea, which is powerful as the state of the sea and ocean–atmospheric linkages are highly latitude dependent.

We conducted a paleohydraulic analysis on storm-generated, hummocky bedforms preserved in marine intervals of the Fountain and Minturn Formations in Pangea’s paleoequator. Reconstructed wave periods and fetch distances indicate the presence of large gravity waves formed under gale-to-hurricane strength winds. These bedforms therefore required cyclonic storms driven by high Coriolis vorticity and high sea surface temperatures. These results are at odds with a cold equatorial hypothesis, and challenge published paleolatitude estimates (<5 degree north) for the Colorado region, suggesting that Pennsylvanian tropical conditions were fundamentally different than previously posited.

Collaborative Development of a Pluvial Flood Model and Adaptation Strategies to Advance Resilience in Baltimore

Ava Spangler^{1,2}, Antonia Hadjimichael^{1,2}

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

²*Earth and Environmental Systems Institute, The Pennsylvania State University, University Park, PA, USA*

Climate change is intensifying the hydrologic cycle, leading to more frequent and severe rainfall-driven (pluvial) flooding in urban areas. In the mid-Atlantic US cities, aging and under-designed stormwater infrastructure is increasingly strained by these events, resulting in recurring damage to property and disruptions to transportation networks. In this study, we combine community engagement with hydrologic modeling to develop and evaluate potential urban flood adaptation strategies. Over a three-year period, local technical experts and community representatives met regularly to discuss flooding concerns, identify priorities, and co-develop adaptation strategies. These discussions informed the development of an urban flooding model (EPA Storm Water Management Model) for the Baltimore Harbor watershed, the focus location of this study. The flooding model integrates complex surface and subsurface stormwater infrastructure data, local expert knowledge, and community insights. We simulate stakeholder-prioritized adaptations, such as green and gray infrastructure strategies. Model results demonstrate that enhanced infrastructure maintenance is the most effective adaptation for reducing flood depths, but has varied effects across the watershed, and can increase flooding in some locations. Spatially concentrated greening provides limited benefit to the watershed as a whole, but moderate benefit in community priority areas. Together, these adaptations have the potential to reduce flood depths by as much as 58% in some locations, greatly reducing property damage and mobility impacts, primary concerns of stakeholders. Future work will implement robust optimization tools to search for adaptations which meet stakeholder objectives and perform highly under varied future climate conditions. This work contributes to the expanding literature on collaborative modeling and demonstrates that community-engaged approaches can enhance model credibility and generate more actionable insights for communities seeking to strengthen climate resilience.

Elevated CO₂ Effects on Water Isotope Fractionation in Trees

Bailey Hoplight¹, Max Lloyd¹

¹*Department of Geosciences, the Pennsylvania State University, University Park, PA, USA*

Plant tissues and their oxygen isotope ($\delta^{18}\text{O}$) compositions are considered valuable records of paleoenvironmental change, capturing shifts in key climate parameters such as temperature, water source, and water availability. Plant tissue $\delta^{18}\text{O}$ represents the combined influence of these processes on the physiochemical response of the plant, and this understanding has been leveraged to reconstruct temperature and hydroclimate conditions across the recent geologic timescale. However, it is important to understand how additional factors could confound this framework. One such factor that remains poorly constrained is the response of plant tissue $\delta^{18}\text{O}$ under varying atmospheric CO₂ concentrations.

This study aims to investigate the effects of elevated CO₂ concentrations on the $\delta^{18}\text{O}$ isotopic composition of plant materials. *Ginkgo biloba* trees were grown in chambers for six growing seasons with CO₂ concentrations ranging from 425 to 1000 ppm. $\delta^{18}\text{O}$ compositions were measured from both bulk wood and cellulose of latewood across multiple annual growth rings. If resulting $\delta^{18}\text{O}$ values are found to be quantitatively related to CO₂ concentrations, then previous and future paleoclimate reconstructions based on the isotopic composition of plant tissues will require careful (re-)calibration to atmospheric CO₂.

This presentation will highlight preliminary results, emerging isotopic trends, and offer initial interpretations as to how these patterns relate to physiological responses of plants under elevated CO₂ conditions.

Orbitrap-MS Characterization of Polycyclic Aromatic Hydrocarbons on the Asteroid Bennu

Mila Matney¹, Kate Freeman¹

¹Department of Geosciences, The Pennsylvania State University, University Park, PA, USA

The return of material from the carbonaceous chondrite asteroid Bennu by the OSIRIS-REx mission provides a unique opportunity to explore pristine organic matter sourced from the origins of the Solar System. Many classes of organic compounds have been identified on Bennu, including bio-relevant molecules like amino acids, nucleobases, and sugars. Aromatic compounds found in carbonaceous chondrites are of particular interest as they are prevalent both in soluble and insoluble organic fractions and have diverse potential origins, ranging from cold interstellar chemistries to high temperature reactions in circumstellar environments in addition to reactions occurring post-accretion on the asteroid parent body. Previous analyses have confirmed the presence of 2-6 ring polycyclic aromatic hydrocarbons (PAHs) on Bennu, with a greater proportion of smaller forms than in studied meteorites. We will further investigate these trends using Orbitrap mass spectrometry (MS) which combines high mass resolution and low detection limits to allow for accurate measurements in sample-limited situations, like that of Bennu. The quantification of major PAH species and overall population of aromatics will be reported on, in the context of other meteorite samples and the history of the Solar System.

Three new Atherospermataceae species from the middle Eocene add 35 million years to the family's Australian history

L. Alejandro Giraldo¹, Peter Wilf¹, Raymond J. Carpenter²

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

²*School of Biological Sciences, Adelaide University, Adelaide, SA, Australia*

The southern sassafrases (Atherospermataceae) are a small plant family of ca. 16 species, ten of which are endemic to Australia. Despite being a dominant element of Australian rainforests today, the earliest continental record of the family dates to the Plio–Pleistocene of Tasmania, thereby limiting interpretations of Atherospermataceae biogeography. Here, we describe three new Atherospermataceae species based on organically preserved leaves from the middle Eocene (ca. 41–38 Ma) Anglesea flora of southeastern Australia, representing the terminal phase of Gondwana. The fossils were examined using light, epifluorescence, and confocal microscopy, and compared with extant material, including all extant Atherospermataceae genera, using the literature, herbarium specimens, cleared-leaf material, and cuticle preparations. Based on leaf architecture and cuticular morphology, two species are assigned to the living genera *Doryphora* and *Laurelia*, while a third, characterized by a densely papillate abaxial epidermis not seen in any living member of the family, is placed in a new fossil genus. These fossils extend the Australian macrofossil record of Atherospermataceae by ca. 35 million years, provide the only pre-Neogene evidence of the family in Australia, and show that Atherospermataceae were already diverse and well-established in southeastern Australia by the middle Eocene, during the final stage of Gondwanan fragmentation. The occurrence of *Laurelia* at Anglesea provides direct evidence that the genus, now extant in New Zealand and southern South America, once extended across a broader range than its modern relicts and prior fossils indicate, and suggests that Australia could have acted as the source or “stepping-stone” region for *Laurelia* in New Zealand.

Thursday 16th April

Poster session 1

1 pm – 2:30 pm

Mingxi Hu

Ph.D. student, 2nd year, Pre-Comps

Decoding the Late Paleozoic Ice Age Paradox: Basin-Scale Filtering of Global Climate Signals

Sabrina Ashik

Ph.D. student, 1st year, Pre-Comps

Leakiness Variability in Arctic River Deltas from SWOT Water Surface Elevation Observations

Ash Keenan

Ph.D. student, 2nd year, Pre-Comps

Morphodynamic Variability Between River Delta Experiments With Identical Input and Boundary Conditions

Laura Kelsall

Ph.D. student, 1st year, Pre-Comps

Microtextural controls on the dissolution of monazite and zircon during biotite-dehydration melting

Drew Moreland

M.S. student, 1st year

Understanding how flow variability and sea level rise impact delta morphodynamics

(continued on next page)

Thursday 16th April

Poster session 1 (continued)

1 pm – 2:30 pm

Doní Anderson

Ph.D. student, 1st year, Pre-Comps

Exploring Deltaic Stratigraphy in Arctic Environments

Edo Nurshal

Ph.D. student, 2nd year, Pre-Comps

Fault Slip and Fluid Flow: Seismic Source Analysis of Slip Patches and Their Role in Fault Permeability

Guilherme Zakarewicz

Ph.D. student, 2nd year, Pre-Comps

DAS–Geophone MASW in Permafrost: Lateral Variability and Implications for Subsurface Characterization in a Changing Arctic

Ula Filipowicz

Ph.D. student, 1st year, Pre-Comps

Modelling the Subduction Interface: The Role of Cohesion in Fault Stability

Zach Baran

Ph.D. student, 2nd year, Pre-Comps

Stratigraphy and Paleohydrology of Fluvial Deposits Near Juventae Chasma in Mars's Southern Highlands

Decoding the Late Paleozoic Ice Age Paradox: Basin-Scale Filtering of Global Climate Signals

Mingxi Hu¹, Elizabeth Hajek¹, Brian Kelley¹

¹Department of Geosciences, The Pennsylvania State University, University Park, PA, USA

The stratigraphic record is our primary archive for understanding Earth's climate history and dynamics, yet sedimentary basins are active filters that "shred," amplify, or distort original climate signals. This challenge is particularly acute in icehouse periods, such as the Late Paleozoic Ice Age (LPIA), regarding its tropical hydroclimate, temperature, and atmospheric circulations. It remains unclear whether these contradictions reflect fundamentally different mechanisms to the Pleistocene, or our inability to read LPIA records across boundary conditions.

I propose to resolve this ambiguity by systematically comparing two equatorial Laurentia basins: the Bird Springs (BS) passive margin and the Midcontinent epeiric sea (MS). While both experienced similar tropical climate, they differed in oceanography, terrestrial flux, and accommodation. By integrating lithostratigraphy, geochemistry (B/Ga for paleosalinity; CIA for weathering), and fauna assemblages, I aim to quantify how basin boundary conditions tuned climate records. I hypothesize that the restricted MS basin amplifies high-frequency, regional "noises" decoupled from global trends, whereas the open-marine BS basin provides a better-fidelity archive for global climate forcing. Further comparison to other LPIA records from various latitudes and locations will test whether LPIA climate patterns (e.g., ITCZ shifts) align with Pleistocene mechanisms. This research establishes sensitivity baselines for comparing local vs. global sedimentary records, providing a critical framework for 1) reconstructing past climate, 2) predicting future climate with geographical differences, and 3) informing resource management.

Leakiness Variability in Arctic River Deltas from SWOT Water Surface Elevation Observations

Sabrina Ashik¹, Anastasia Piliouras¹

¹Department of Geosciences, The Pennsylvania State University, University Park, PA, USA

Arctic river deltas play a critical role in regulating the delivery of freshwater, sediment, and nutrients to the coastal ocean, influencing sea ice dynamics and coastal ecosystems. Discharge and sediment concentrations are often measured upstream at the delta apex, and how these fluxes are partitioned and redistributed across distributary networks remains poorly understood. Flow partitioning within these networks depends on their hydraulic structure, including the degree of lateral exchange between channels and adjacent islands. One metric used to quantify this behavior is channel-island “leakiness,” formulated by Hines and Piliouras (2024), which represents the likelihood of water leaving the channel as a function of channel geometry and water surface gradient. Leakiness has been calculated for the Mackenzie Delta using historical water level and discharge data from gauge stations; however, its variability across Arctic systems remains poorly constrained at basin scales due to sparse in situ observations. In this study, we quantify temporal and spatial variability in channel-network leakiness across major Arctic deltas using remotely sensed Water Surface Elevation (WSE) data from the Surface Water and Ocean Topography (SWOT) mission.

We extracted SWOT WSE time series at apex and outlet locations from August 2023 to January 2026, used WSE as a proxy for water level, and calculated monthly leakiness values. We then evaluated the relationship between leakiness and discharge using the framework proposed by Hines and Piliouras (2024). Preliminary results indicate a consistent positive relationship between discharge and leakiness at both apexes and outlets, with stronger sensitivity at the apex locations than at the outlets. However, we observed a large inter-delta difference, suggesting that leakiness-discharge relationships derived from a single system cannot be generalized across Arctic deltas. These findings advance our understanding of flow partitioning in Arctic distributary networks and establish a foundation for quantifying large-scale hydraulic connectivity from satellite observations.

Morphodynamic Variability Between River Delta Experiments With Identical Input and Boundary Conditions

Ash Keenan¹, Anastasia Piliouras¹

¹Department of Geosciences, the Pennsylvania State University, University Park, PA, USA

Flume experiments are valuable tools for simulating and studying complex sediment transport and morphodynamic processes that are difficult to observe in natural environments. However, these experiments are resource-intensive and thus rarely repeated, raising questions about the reproducibility of results. This research utilizes a series of five identical experiments (Su1–Su5) repeated under the same input and boundary conditions in order to investigate the inherent variability between physical delta experiments. We examine variations in delta characteristics such as area growth, wetted frequency, perimeter/area ratio, shoreline roughness, and fractal dimension using overhead photography and 3D photogrammetry. Preliminary results show that area growth rates remain positive and approximately linear across all trials, reflecting constant progradation under uniform water and sediment input, though final delta sizes diverge slightly. Wetted fraction values are comparable among experiments (means ~0.40–0.55), indicating occupation of a similar partially channelized regime. Perimeter/area ratio exhibits a consistent downward trend as deltas expand, with small fluctuations driven by planform differences. In contrast, shoreline roughness reveals meaningful divergence: Su3 experienced a major late-stage channel avulsion that produced fewer, larger distributaries and a sustained increase in shoreline complexity relative to the other experiments, which were characterized by gradual channel migration and numerous smaller outlets. Fractal dimension captures greater similarity of the shoreline overall but also reflects elevated complexity in Experiment 3 following avulsion. Ongoing work will incorporate earlier experimental hours and analyze photogrammetry-derived digital elevation models to assess differences in slope and channel relief. Together, these results demonstrate that while many bulk morphodynamic metrics are statistically similar under identical forcing, stochastic events can generate measurable divergence, underscoring the importance of replication in experimental geomorphology.

Microtextural controls on the dissolution of monazite and zircon during biotite-dehydration melting

Laura Kelsall¹, Peter Kelemen², Andrew Smye¹

¹ *Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

² *Columbia University, Lamont Doherty Earth Observatory, Palisades, NY, USA*

Differentiation of the continental crust into an upper layer enriched in heat-producing elements (U, Th, K) and a complementary lower crust depleted in these elements is key for continental stability (e.g., Mareschal & Jaupart, 2013; Sandiford et al., 2002). Recent analysis indicates that ultrahigh temperatures (>900°C) may be required to effectively redistribute U and Th from the lower to upper crust (Smye & Kelemen 2025). Such temperature requirements are at odds with equilibrium thermodynamic modelling (Yakymchuk & Brown, 2014), and so the exact mechanism that accounts for this observation is uncertain.

Here we test the hypothesis that mineral armouring modulates the dissolution of monazite and zircon—key Th and U carrier phases, respectively—in silicate melt produced during biotite-dehydration melting. We present microtextural and trace element geochemical observations from a suite of aluminous metasedimentary rocks (Ivrea-Verbano Zone) that preserve a record of partial melting under peak temperatures spanning the biotite-dehydration melting interval (700-900°C). Automated SEM mapping of thin sections reveals a strong positive correlation between biotite and monazite modal abundances. In the amphibolite-facies samples 8-27% of monazite is either included or adjacent to biotite grains whereas in granulite-facies and ultrahigh temperature samples that decreases to 0-22% and 0% respectively. The amount of monazite included in garnet increases from 3-10% in amphibolite to 75% in ultrahigh temperature samples. Zircon abundance is poorly correlated to biotite but still shows a similar trend of decreasing association with biotite (15-44% to 0%) and increasing association with garnet (3-29% to 38%) from amphibolite to ultrahigh temperature samples.

These findings suggest that the textural availability of accessory phases imposes an important control on the flux of U and Th into silicate melt produced during anatexis. Prior to the consumption of biotite, included carrier phases may be shielded from interacting with melt; peritectic growth of garnet serves to sequester carrier phases in the residuum. Work is ongoing to incorporate this microtextural framework into a forward model describing the evolution of melt U, Th concentrations during biotite-dehydration melting and to explore implications for crustal differentiation.

Understanding how flow variability and sea level rise impact delta morphodynamics

Andrew Moreland¹, Liz Hajek¹, Anastasia Piliouras¹

¹*Department of Geosciences, The Pennsylvania State University, University Park PA, USA*

Global temperatures are rising, accelerating sea-level rise that threatens the stability of deltaic environments. More than 300 million people live in or near vulnerable delta systems, many of which already experience land loss, subsidence, and amplified flood hazards. Global warming is also influencing river discharge patterns, which can trigger more abrupt avulsions and potentially deadly floods.

Most existing studies have examined sea-level rise and flow variability independently. However, these processes often operate simultaneously in deltas, meaning their combined effects on channel mobility, sediment distribution, and avulsion behavior remain poorly constrained. This project aims to address this gap by experimentally investigating how sea level rise and discharge variability affect delta dynamics both individually and together.

My experiments will be conducted in the Sediment and Natural Dynamics Basin for Observation and eXperiments (SANDBOX), a controlled physical basin that allows direct manipulation of sea-level rise and discharge variability. Over the course of my M.S. research, I will conduct approximately nine experiments looking at how sea level rise and discharge variability will affect channel mobility. Stratigraphic analyses will follow to quantify differences in sediment deposition and avulsion signatures across runs.

My hypothesis is that rising sea levels and increased flow variability will accelerate shoreline change and increase avulsion frequency in deltaic systems. From this we will learn more about stratigraphic stacking patterns and have a better understanding of accommodation and sediment supply. By linking sea-level forcing with flow variability, it fills a key gap in geomorphology for interpreting natural deltas. The broader significance lies in improving hazard assessment and management strategies for vulnerable coastal regions.

Exploring Deltaic Stratigraphy in Arctic Environments

Doni Anderson¹, Anastasia Piliouras¹, Benjamin Cardenas¹

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

River deltas leave deposits that hold a useful record of past environmental conditions. Arctic deltas on Earth provide insights about how ice cover and permafrost affects depositional patterns, potentially holding clues to constrain identification of ancient deltas that may have formed under the influence of ice. Despite its current presence and abundance, deltaic stratigraphy in Arctic environments has not been heavily explored, limiting our understanding of ancient deltas. We aim to fill this gap.

We used numerical models of delta morphodynamics in Arctic conditions to examine how permafrost and ice impact delta stratigraphy. We used stratal volumes produced by these models to explore differences between permafrost at varying erodibilities and different ice thicknesses (no permafrost to 65% permafrost erodibility and no ice to 4.5m ice thickness), and to quantify sand fraction throughout the delta. Preliminary results show that ice and permafrost have a noticeable impact on delta formation. As ice thickness increases and permafrost erodibility decreases, there is an increase in mud in the delta plain, likely resulting from sand being bypassed out of the subaerial system because the permafrost inhibits accommodation for the sediment supply. Sand fraction generally decreases down delta, but the combination of thicker ice and lower permafrost erodibility results in a shallower decrease in sand fraction in comparison with thinner ice/higher permafrost erodibility or no ice/no permafrost model run outputs. When ice thickness decreases and permafrost erodibility increases, there is more mixing of the mud and sand sediment throughout the delta, resulting in a higher sand fraction overall. Future analyses will describe channel patterns by quantifying channel lateral mobility and vertical stacking and measuring channel geometry. This work will continue to explore the implications of these findings for connections between ancient delta stratigraphy and paleoclimate. These first steps could prove useful for ancient delta studies on places such as Mars where its paleoclimate is still uncertain.

Fault Slip and Fluid Flow: Seismic Source Analysis of Slip Patches and Their Role in Fault Permeability

M. Edo Nurshal¹, Junpeng Wang², Pengliang Yu^{1,2}, Tushar Mittal¹, Derek Elsworth^{1,2}

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

²*Department of Energy and Mineral Engineering, The Pennsylvania State University, University Park, PA, USA*

Fluid injections can trigger fault reactivation, microearthquakes (MEQs), and permeability changes that are critical for geothermal energy production and subsurface waste disposal. Recent studies have shown that fault permeability can be quantitatively predicted from the cumulative seismic moment released by MEQs. To investigate the physical mechanisms behind this relationship, we conducted laboratory fault reactivation experiments on a Utah FORGE granitoid sample and analyzed acoustic emission (AE) signals generated during stepwise increases in injection pressure. Frequency analysis of thousands of calibrated AE signals reveals that fault reactivation occurs through the activation of numerous AE source patches with millimeter-scale radii, far smaller than the fault surface of the sample. The cumulative area of these reactivated patches exceeds the fault area by approximately one to two orders of magnitude ($\sim 10\text{--}50\times$) during each pressure step. These results demonstrate that permeability enhancement is not driven by a single large slip event but emerges from the sequential activation and interaction of multiple slip patches that progressively establish a connected flow pathway.

DAS–Geophone MASW in Permafrost: Lateral Variability and Implications for Subsurface Characterization in a Changing Arctic

Guilherme Zakarewicz¹, Gabriel Rocha dos Santos¹, Tieyuan Zhu¹, Ming Xiao², Eileen R. Martin³, Ziyi Wang⁴

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

²*Department of Civil and Environmental Engineering, The Pennsylvania State University, State College, PA, USA*

³*Department of Geophysics, Department of Applied Math and Statistics, Colorado School of Mines, Golden, CO, USA*

⁴*Department of Civil and Environmental Engineering, Colorado School of Mines, Golden, CO, USA*

Permafrost is a subsurface condition where soil or rock remains below freezing temperatures for at least two consecutive years, covering large portions of the Northern hemisphere and Alaska. However, recent climate changes are accelerating its degradation, resulting in subsidence, riverbank erosion, and infrastructure damage. Detecting and tracking permafrost change remains challenging because many traditional observations are expensive and invasive. In this context, geophysical methods offer a practical complement by identifying differences in the physical properties. For instance, elastic seismic wave speeds respond strongly to variations in ice and liquid water content.

In this study we compare Vs profiles derived from (i) a distributed acoustic sensing (DAS) fiber and (ii) conventional geophone arrays at five different locations in Utqiagvik, Alaska. Since the DAS and geophones were not exactly co-located (5-45 meters offset), the comparison provides enough lateral separation to test spatial variability. Although some sites presented similar Vs profiles, others showed larger discrepancies, exceeding the uncertainty bounds (typical ΔV_s of ~100–400 m/s). We interpret those differences as reflecting lateral heterogeneity at dozen-of-meters scales rather than differences in the acquisition systems. To provide a quantitative example of a physical parameter that could illustrate the observed differences, we apply a rock-physics relation to estimate ice/water volumetric content and quantify how much variability in water saturation would be required to explain the observed Vs differences (typical ΔS_w of ~0.15). We also compare the Vs-derived saturation profiles with temperature-based estimates from power-law relationships reported by an independent study to check plausibility. Our results show that permafrost should not be treated as laterally uniform when constructing velocity models and interpreting time-lapse changes, especially in coastal polygonal tundra where we expect variations in ice content, salinity and porosity. Furthermore, spatially dense measurements are essential for interpreting apparent changes in regions with complex cryostructure and hydrology.

Modelling the Subduction Interface: The Role of Cohesion in Fault Stability

Urszula Filipowicz¹, Donald Fisher¹, John Hooker², Tushar Mittal¹

¹*Department of Geosciences, The Pennsylvania State University*

²*Department of Atmospheric Science, Environmental Sciences, University of Incarnate Word*

Earthquakes are among the most significant geohazards on Earth, with the largest events occurring along subduction zones. The mechanisms responsible for unstable slip along megathrust faults are believed to be primarily controlled by friction, normal stress, and fluid pressure. These concepts are reflected by earthquake models, which typically assume slip follows the rate-and-state friction laws, where friction evolves in response to velocity stepping, but cohesion is not included in the calculations. More recent experimental work, described by the CNS model, has shown that cohesion might be an important aspect to consider when determining the type of failure that will occur along the subduction interface, highlighting how behaviour of faults can be defined from grain-scale mechanisms. Based on this microphysical research whilst also challenging the current paradigm, the hypothesis of this study is that cohesion variations can trigger fault instabilities independent of velocity weakening.

In order to test the role of cohesion, a numerical framework called the MEFISTO (*Mineralization, Earthquake, and Fluid flow Integrated SimulaTOR*) model will be used. By simplifying the problem to a zero-dimensional block, the separated effects of cohesion and friction can be examined and quantified. This will improve MEFISTO's ability to simulate earthquake nucleation and strengthening, allowing predictions for how different variables such as temperature and strain rates during interseismic periods affect cohesion, which can impact slip behaviour and result in earthquakes. Once cohesion is incorporated into the full MEFISTO framework, the resulting forecasts of earthquake size and distribution will be compared with the Global Centroid Moment Tensor (GCMT) catalogue to evaluate the predictions of the model, and hence, the significance of cohesion in fault stability. Such results would develop understanding of subduction zones and guide future applications of the model directly to field observations and laboratory analysis.

Stratigraphy and Paleohydrology of Fluvial Deposits Near Juventae Chasma in Mars's Southern Highlands

Zachary J. Baran¹, Benjamin T. Cardenas¹

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

Sedimentary basin stratigraphy records information about a region's past depositional environments. Ancient fluvial deposits on Mars are often found within regional or large-scale topographic lows like craters or along the margin of the northern lowlands. However, a recent study showed that abundant fluvial strata also exist in the southern highlands, suggesting that stratigraphic basins also exist in locations incongruent with the modern topography. One of these southern highland strata locations is Juventae Plateau (62.5°W 4.50°S). Previous studies have identified hydrated sulfate minerals in strata across the plateau, as well as ancient river channel deposits, suggesting past water, but stratigraphic relationships have not been quantified and a depositional setting to explain both features has not been put forward.

We used high-resolution satellite images and digital elevation models to map ridges previously interpreted to be exhumed river-channel belts based on exposed point-bar structures and stacking patterns similar to exhumed channel belts on Earth and elsewhere on Mars. We measured the thicknesses of layered beds and ridges and found that ridges are consistently thicker than the layered deposits. The total elevation range occupied by the ridges and layered deposits suggests over 250 meters of basin fill. We measured orientations of strata and found both relatively flat-lying beds and inclined, truncated strata, which we interpret as fluvial floodplain and bar deposits respectively. Dip direction and exposed point-bar structures suggest a paleoflow direction of north-to-south, which is notably opposite to the modern regional topographic slope.

We measured ridge caprock thicknesses to reconstruct paleochannel depths and to look for downstream changes in paleochannel depth, testing hypotheses for deltaic (downstream thickening) and terminal fan (downstream thinning) channels, further constraining the depositional setting. Reconstructed channel depths were also used to better understand water availability and climate during basin filling by reconstructing water discharge using established relationships.

Thursday 16th April

Oral session 3

3:00 pm – 4:00 pm

3:00 pm – 3:15 am

Jasmine Walker

Ph.D. student, 4th year, Post-Comps

The Role of Fine Sediment in Channel Mobility & Stratigraphic Architecture

3:15 pm – 3:30 pm

Ran He

Ph.D. student, 5th year, Post-Comps

Iron carbonate as a robust archive for reducing paleoenvironmental conditions: oxygen and clumped isotopes evidence from burial diagenesis

3:30 – 3:45 pm

Yusuke Kubota

Ph.D. student, 4th year, Post-Comps

The role of the mantle decompaction layer in Hadean volcanism

3:45 am – 4:00 pm

Oliver Neilson

Ph.D. student, 3rd year, Post-Comps

Quantifying the timing and volume of volcanic degassing in deep time: A case study from the Sinemurian – Pliensbachian Boundary Event

The Role of Fine Sediment in Channel Mobility & Stratigraphic Architecture

Jasmine Walker¹

¹Department of Geosciences, The Pennsylvania State University, University Park, PA, USA

Understanding how fluvial systems respond to change and record the history of landscape changes is essential for sustainably managing ecosystems, remediating environmental contaminants, efficiently exploring for natural resources, and successfully mitigating risks associated with river movements and floodplain inundation. Rivers play a key role in shaping the modern landscapes we inhabit and the buried deposits that form stratigraphy, from which we interpret the history of surface conditions on Earth. The integrated effect of silts and clays on fluvial landscape dynamics is recorded in the stratigraphic architecture of ancient river deposits and can enhance or occlude fluid flow in buried aquifers and reservoirs. Understanding how fine-sediment supply impacts channel mobility, stratigraphic architecture, and subsurface fluid flow is, therefore, key for reconstructing paleo-landscape conditions and predicting the distribution and connectivity of subsurface reservoirs, aquifers, and potential geothermal and carbon storage sites.

A key challenge in understanding the relationship between the supply of fine sediment and resulting stratigraphic architecture is the interplay between processes occurring at both spatially and temporally variable scales. I aim to close this gap through inquiry into modern rivers and ancient river deposits that assess the role of fine sediment in channel mobility and long-term channel stratigraphic architecture, or “stacking”, in sedimentary basins. Using a combination of remote sensing of modern rivers, subsurface (seismic and well log) observations of ancient river deposits, and numerical modeling, I propose to answer the following science questions: (1) How does suspended-sediment supply impact channel mobility in global alluvial rivers? (2) What is the relationship between fine-sediment supply, channel mobility, and channel-belt depositional architecture in fluvio-deltaic settings?

Iron carbonate as a robust archive for reducing paleoenvironmental conditions: oxygen and clumped isotopes evidence from burial diagenesis

Ran He¹, Max K. Lloyd¹, Brian Currie², Miquela Ingalls¹

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

²*Department of Geology and Environmental Earth Science, Miami University, Oxford, OH, USA*

Siderite (FeCO_3) is a redox-sensitive carbonate mineral that is pervasive in the rock record and can enable reconstructions of ancient reducing environments on Earth and Mars. However, post-depositional processes like burial diagenesis and surficial weathering can alter primary isotopic signatures through dissolution and reprecipitation ($\delta^{13}\text{C}$, $\delta^{18}\text{O}$, and Δ_{47}), limiting the reliability of these proxies. Unlike calcite and dolomite, the susceptibility of isotopic clumping in siderite to overprinting by these processes is poorly understood. We hypothesize that siderite is more resistant to these processes than calcite during fluid-mediated diagenesis due to its lower solubility. To test this hypothesis, we analyzed siderite and calcite samples from the Paleocene Fort Union Formation in the Wind River Basin, Wyoming. We collected coeval terrestrial siderite concretions and cement from two outcrops and six drill cores spanning a range of maximum burial depths (1.5-3.5km) due to tectonics, allowing us to distinguish between water-rock alteration during burial diagenesis and surficial weathering.

All calcite samples yield elevated $T(\Delta_{47})$ temperatures ($44.5 \pm 1.2^\circ\text{C}$ to $55.9 \pm 2.6^\circ\text{C}$) paired with low and variable $\delta^{18}\text{O}$ values (-18.9 to -11.2‰), whereas siderite from outcrops and shallow cores records plausible Earth surface temperatures ($9.1^{+5.7}_{-5.4}^\circ\text{C}$ to $27.6^{+5}_{-4.8}^\circ\text{C}$). Two micritic siderite Δ_{47} from deepest core (3.5 km burial) however, shows elevated $T(\Delta_{47})$ temperatures ($43.7 \pm 0.9^\circ\text{C}$ to $60.2^{+4.9}_{-4.7}^\circ\text{C}$), indicating resetting through recrystallization. Of these two altered siderite samples, one exhibits textural and chemical evidence of recrystallization during burial under scanning electron microscopy, while the other shows no petrographic signs of alteration. Combining $T(\Delta_{47})$ and $\delta^{18}\text{O}$, water-rock interaction model suggests the altered siderite has experienced rock-buffered diagenesis (water-to-rock ratio < 0.5), whereas calcite has experienced fluid-buffered diagenesis (water-to-rock ratio > 1). Together, these results demonstrate that siderite $\delta^{18}\text{O}$ and $T(\Delta_{47})$ are more robust to burial alteration than calcite, highlighting the potential of siderite as a reliable proxy for ancient reducing aqueous environments.

The role of the mantle decompaction layer in Hadean volcanism

Yusuke Kubota¹, John Rudge², Bradford Foley¹

¹*Department of Geosciences, Pennsylvania State University*

²*Department of Earth Sciences, University of Cambridge*

How the very early Earth lost its internal heat remains a subject of debate. Early Earth may have been characterized by extensive magmatism due to a hot mantle, which then acted as the primary heat loss mechanism, or been more volcanically quiescent, where heat conduction through the lithosphere served as the primary heat loss mechanism. The primary mode of early Earth heat loss would then strongly influence tectonics and crust formation, the long-term thermal evolution of the interior, and surface environments where life could originate in the Hadean or Eoarchean.

Our recent crustal evolution model suggests that mantle melt production and mafic extrusive volcanism must have been limited prior to 3.6 Ga to remain consistent with Hf isotope data. We hypothesized that these geochemical constraints require a 'quiescent Earth' with a low melt production rate (<0.6 mm/yr). However, the actual magma supply is governed by complex geodynamic factors: specifically, the mechanics of melt generation, ascent, and accumulation at the mantle-crust boundary. Understanding these physical mechanisms is critical, particularly when evaluating high-flux regimes such as heat-pipe tectonics, which may be incompatible with the low rates inferred from the geochemical record.

A critical phenomenon affecting this supply is the formation of a decompaction layer beneath the mantle-crust interface. Since the crust acts as a rigid thermal boundary, temperatures drop rapidly near this interface. Consequently, ascending melt encounters a freezing horizon that acts as a permeability barrier, causing it to accumulate. Within this zone, the decompaction layer and accumulated magma generate significant melt overpressure relative to the solid matrix, driving magma into the plumbing system and initiating ascent. Therefore, characterizing the dynamics of the decompaction layer is crucial for understanding the physical controls on melt supply.

Recent numerical modeling of Io, an active heat-pipe body, demonstrates that crustal thermal structure is controlled by the physics of two-phase melt transport. This model suggests that magma transport is driven by mantle overpressure at the decompaction layer but limited by solidification within the plumbing system. Applying this physical framework to the Hadean, we tested which mantle dynamics and temperature ranges are compatible with the restricted melt fluxes required by the geochemical record. Preliminary results demonstrate the existence of a decompaction layer, where both effective pressure and extraction rates increase significantly with porosity. By systematically exploring the parameter space, we identify the specific mantle geodynamic conditions required to align plateau melting tectonics with the Hf isotope constraints.

Quantifying the timing and volume of volcanic degassing in deep time: A case study from the Sinemurian – Pliensbachian Boundary Event

Oliver Neilson¹, Isabel M. Fendley¹

¹*Pennsylvania State University, State College, PA, USA*

Understanding deep-time climatic perturbations relies on quantifying the initial drivers them. Earth system perturbations are highly sensitive to, among other parameters, the timing and duration of volcanic degassing. Currently, these parameters are coarsely constrained, with volatile estimates coming from melt inclusion data and radiometric dating¹. However, recent work has highlighted the power of combining sedimentary mercury (Hg), a volcanic tracer, and Hg cycle box models to estimate the tempo and volume of volcanic degassing in deep time².

Here, we present a quantitative high-resolution degassing history through the Sinemurian - Pliensbachian Boundary Event (SPBE). This protracted negative “U-shaped” carbon isotope excursion lasted for over 4 million years in the Early Jurassic (ca. 190 Ma). We utilise over 1600 samples collected from the recently drilled core at Prees, Cheshire Basin, U.K., as part of the International Continental Scientific Drilling Program JET project.

The SPBE is broadly coeval with increased rifting from the break-up of Pangea, the opening of the Hispanic Seaway, and a pulse of volcanic activity from the Central Atlantic Magmatic Province^{3–5}, all of which may have contributed to its shape and duration. We quantify the tempo and volume of volcanic degassing during the SPBE using a geochemical machine-learning framework to isolate volcanically sourced Hg, followed by identification of the best-fit degassing scenarios using a global Hg box model. To see how this would affect the carbon isotope record, we use the best-fit degassing scenario as an input to a carbon cycle model. We compare the modelled record to the observed record and discuss major differences.

This novel method allows for a clearer picture of past climatic feedbacks and Earth system dynamics; quantifying past climatic changes and their drivers are essential for better contextualising modern and future climate change.

Friday 17th April

Poster session 2

9:00 am – 10:30 pm

Sam Williams

M.S. student, 1st year

Quantifying sediment storage and evacuation in response to compound earthquake and typhoon hazards in the Liwu River, Taiwan

Chandan Sahu

Ph.D. student, 2nd year, Pre-Comps

How does composition variation affect mantle solidus and thermal evolution of rocky planets?

Christian Lundy

Ph.D. student, 2nd year, Pre-Comps

The Stabilization of Wastewater-Derived Phosphorus in a Florida Keys, USA Karstic Aquifer

Anthony Delgadillo Salas de la Tierra

Ph.D. student, 1st year, Pre-Comps

Reconstructing Late Pleistocene/Holocene Fire–Vegetation–Human Feedback in the Lake Malawi Basin Using Core MAL05-2A

Olivia DiPrinzio

Ph.D. student, 1st year, Pre-Comps

What Factors Drive Farmers' Climate Risk Perceptions? A case study in Chitwan, Nepal

Zach Loveall

M.S. student, 1st year

Dual-Clumped Isotope Disequilibrium in Terrestrial Carbonates

(continued on next page)

Friday 17th April

Poster session 2 (continued)

9:00 am – 10:30 pm

Lara Williams

M.S. student, 1st year

Investigating deep critical zone structure using landslide scars in the Taiwan Central Range

Tsunami Sharmba

Ph.D. student, 1st year, Pre-Comps

A Network Analysis of Administrative Calls and User Interdependencies in the Colorado River Basin

Jackson Saftner

Ph.D. student, 4th year, Post-Comps

Low-Cost Monitoring of Inert Gas Stimulation with DAS

Gabriel Dos Santos

Ph.D. student, 5th year, Post-Comps

Distributed Sensing of Frost Quakes and Ice-Wedge Cracking in Utqiagvik, Alaska

Dayang Zhou

Ph.D. student, 2nd year, Pre-Comps

Listening to Arctic Sea Ice: Seismic Tremors as a Proxy for Ice Strength

Young Kim

Ph.D. student, 5th year, Post-Comps

Assessing Post-Collapse Flank Instability at Anak Krakatau Using Satellite Remote Sensing

Quantifying sediment storage and evacuation in response to compound earthquake and typhoon hazards in the Liwu River, Taiwan

Samuel Williams¹, Eric Kirby¹, Roman DiBiase¹

¹Department of Geosciences, The Pennsylvania State University, University Park, PA, USA

In steep, tectonically active landscapes, earthquakes and precipitation can generate large, episodic sediment pulses that transiently alter channel morphology. The timescales over which these pulses are evacuated, and the sensitivity of transport to large discharge events, remain uncertain. Understanding how sediment pulses propagate through boulder-dominated rivers is necessary for predicting the long-term evolution of active mountain belts. Because these transient responses influence channel longitudinal profiles and the pace at which rivers adjust to tectonic forcing, constraints on sediment storage and evacuation are essential for linking short-term events to long-term landscape evolution. A natural experiment is posed by the Liwu River following the 2024 Hualien earthquake in the Central Range of Taiwan. This Mw 7.2 event triggered widespread co-seismic landsliding, and an intense subsequent typhoon season had multiple storms capable of triggering additional landsliding and mobilizing stored sediment. I will quantify spatial and temporal patterns of sediment storage and evacuation in the Liwu River using a multi-temporal dataset to test the following hypotheses: (1) sediment aggradation will be clustered downstream of major landslide and tributary inputs rather than uniformly distributed along the reach, and (2) reaches with greater pre-event boulder cover will retain sediment longer during floods.

This study leverages a high-resolution Unmanned Aerial Vehicle (UAV) survey conducted in 2023 that provides centimeter-scale orthophotos and 3D point clouds for baseline channel morphology. These data will be integrated with a time series of 10–50 cm scale resolution orthophotos and a repeat UAV survey to quantify bed elevation change, sediment storage, and boulder exposure from 2023-2026 across a ~4 km reach of the Liwu River. Together, these observations will provide new constraints on sediment residence time and the role of boulder cover in controlling sediment transport following extreme disturbances.

How does composition variation affect mantle solidus and thermal evolution of rocky planets?

Chandan Sahu¹, Bradford Foley¹, Aaron Wolf²

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

²*SETI Institute, Mountain View, CA, USA*

Understanding exoplanet interiors remains a challenge due to the inherent difficulty of direct observation. While their atmospheres are observed using advanced telescopes, interior modeling provides information about their interior structure, composition, and evolution relevant for interpreting atmospheric observations. Current models often assume an Earth-like mineralogy, however rocky exoplanets can come in ranges of different compositions. Studies from the Hypatia catalog and other stellar composition catalogues show that stars can have a wide range of elemental abundances, leading to diverse mineral compositions in terrestrial planets, as they reflect the refractory materials present in the host star.

Our research investigates how variations in interior composition affect the mantle mineralogy and solidus, the viscosity and consequently the planet's thermal history. The solidus, marking the onset of melting, is crucial for heat and melt transport. On Earth, olivine and pyroxene are the primary minerals in the upper mantle. Variability in elemental abundances in stars suggests planets may have diverse olivine-pyroxene (Ol-Px) ratios. We use the MELTS model to determine how different mineralogies affect the mantle's solidus profile. Our results indicate higher solidus temperatures for higher Ol-Px ratio. We then integrate the calculated solidii with the mantle adiabat and heat sources/sinks to simulate interactions between mantle composition, temperature, viscosity, and lid growth over geological timescales. We also model the viscosity using flow laws and creep mechanisms, to observe the behaviour with changing Ol-Px ratio. This helps us understand melt production, outgassing, volcanism, and other dynamic processes during a rocky planet's evolution, aiding in predicting the thermal and geological history of exoplanets. Understanding these is essential for identifying the presence of tectonic activity and volcanism, and can create and sustain habitable conditions.

The Stabilization of Wastewater-Derived Phosphorus in a Florida Keys, USA Karstic Aquifer

Christian Lundy¹, Eleanor Spielman-Sun², Lee Kump¹, Miquela Ingalls¹, and Rachel Housego¹

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

²*SLAC National Accelerator Laboratory, 2575 Sand Hill Road, Menlo Park, CA, USA*

Submarine groundwater discharge is an often overlooked source of nutrient loading in coastal ecosystems. Phosphorus (P) inputs into ocean surface waters can cause eutrophication, leading to algal blooms and deoxygenation. Shallow wastewater injection in Marathon, FL injects up to 700,000 liters of high-P treated wastewater daily approximately 30 meters into the Key Largo Limestone (KLL) aquifer for disposal. P concentrations are greatly decreased within the plume due to dilution, adsorption of P to the calcite lattice, and precipitation of P minerals. This P sink is under threat as rising sea levels cause saline intrusion into previously fresh aquifer regions, leading to P desorption and possible P mineral dissolution. This study aims to characterize the P minerals precipitated from KLL shallow wastewater injection to assess P sink stability under future saline intrusion scenarios.

Carbonate cores were collected at three locations along a cross-island transect through the Marathon wastewater plume in September 2025. Adjacent monitoring wells were instrumented with conductivity-temperature-depth sensors to determine core salinity zones and water samples were collected for plume geochemical characterization. Thin sections from the cores were selected from fresh and saline aquifer regions and sections of varying secondary porosity. They will be analyzed using μ -X-Ray Absorbance Near Edge Structures (μ -XANES) and μ -X-Ray Fluorescence (μ -XRF) at the Stanford Synchrotron Radiation Lightsource (SSRL) and Scanning Electron Microscopy/Energy Dispersive Spectroscopy (SEM/EDS) at the Penn State Materials Characterization Laboratory to determine P mineral speciation. We hypothesize that more stable P minerals will be present in aquifer regions with lower salinity and less secondary porosity, as decreased groundwater residence time from preferential karstic pathways leads to decreased P mineral precipitation. This project holds important implications for P loading under future saline intrusion scenarios and can provide important geochemical constraints for reactive transport modeling of shallow wastewater injection.

Reconstructing Late Pleistocene/Holocene Fire–Vegetation–Human Feedback in the Lake Malawi Basin Using Core MAL05-2A

Anthony Delgadillo Salas de la Tierra¹, Sarah Ivory¹

¹Department of Geosciences, The Pennsylvania State University, University Park, PA, USA

Lake Malawi is an important sedimentary archive for understanding the long-term changes between tropical African climate, ecosystems, and early human activity. During the Pleistocene, East Africa experienced megadroughts causing permanent ecological reorganization between ~85-70 ka, shifting from Afromontane forests to miombo woodlands. It is unclear if the transition is solely a climate shift or if early anthropogenic fire-vegetation feedback occurred. Addressing this gap has been challenging due to chronological uncertainties and coarse sampling resolutions during the Late Pleistocene-Holocene transition. This project tests the hypothesis that early human fire use during the 85-70 ka transition created self-sustained ecological feedback that later intensified during the Holocene agropastoralism.

This project aims to analyze 56 sediment samples from core MAL05-2A, spanning the last ~123,000 years with some intervals studied at a higher resolution such as a millennial scale. By utilizing pollen analysis to track vegetation changes with microscopic charcoal analysis to measure fire activity, this study will develop an improved understanding of environmental conditions in southeast Africa. Results are expected to reveal two major phases of fire intensification, the ~85-70ka transition and the Holocene. This fire intensification is suspected to be characterized by a continuous directional shift toward grass dominated savanna assemblages. This research aims to determine whether fire-maintained landscapes in Southeast Africa are a modern occurrence or a legacy of Pleistocene anthropogenic influence, providing a baseline for understanding modern landscape degradation and ecosystem resilience under climate change.

What Factors Drive Farmers' Climate Risk Perceptions? A case study in Chitwan, Nepal

Olivia DiPrinzio^{1,2}, Nicolas Choquette-Levy^{1,2}

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

²*Earth and Environmental Systems Institute, The Pennsylvania State University, University Park, PA, USA*

Under climate change, global crop yields are projected to decline significantly, with staple crops showing reductions of up to 10% per degree of warming (°C). Particularly, this reality is most impactful on socio-economically vulnerable communities, such as farmers in South Asia, whose incomes are predicted to decline by 28% annually on average due to climate change's impacts. Concurrently, climate and agricultural policies worsen existing adverse circumstances by omitting community climate risk perceptions. Such policy frameworks fail to integrate community-specific risks, tolerance, and exposure, thereby threatening livelihoods reliant on agriculture. To date, there is a limited understanding of the effectiveness of risk-informed policies under climate change uncertainty, whether in standalone analyses or compared with non-risk-informed policies.

To address this gap, multivariate statistical methods were employed to examine the drivers of farmers' hazard risk perception in Chitwan, one of Nepal's most climate-vulnerable regions, using a 2022 survey. Guiding this process, causal modeling identified three key feedback mechanisms hypothesized to govern farmers' reported exposure: insurance levels, remittance income, and perspectives on farming as a livelihood strategy. Initial statistical analysis demonstrates the importance of groundwater and irrigation accessibility, the varied influence of demographic variables, and remittances' minimal influence on the frequency of self-reported exposure to climate-influenced hazards, the last of which contradicts the proposed hypothesis of remittances' impact. Our results demonstrate inherent variability in hazard perceptions, such that while certain drivers are relevant across all hazards, a more tailored approach to driver consideration is required. Consequently, a signal exists for hazard-specific policies, such as hazard-specific vulnerability indices. As these are initial results, next steps include statistical analysis of farmers' perceptions of future risks, time-series analysis to provide a more nuanced view of variability in perceptions, self-reports compared against climatological data (e.g. SPEI), and an exploration of hazard-specific exposure policies.

Dual-Clumped Isotope Disequilibrium in Terrestrial Carbonates

Zachary Loveall¹, Ellen Olsen¹, Miquela Ingalls¹

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

Terrestrial carbonates preserve valuable paleoclimate records in their isotopic compositions. However, kinetic effects associated with dissolved inorganic carbon cycling, including CO₂ degassing, absorption, and water mixing can shift isotopic systems out of equilibrium. Because isotopic equilibration in the DIC–H₂O system occurs over hours to days, rapidly forming carbonates may preserve transient disequilibrium signals, complicating paleoclimate records. Carbonates from the Green River Formation of Wyoming and Utah record lacustrine, wetland, and shoreline environments during the Early Eocene Climatic Optimum, a warm interval characterized by elevated pCO₂ and high temperatures. These deposits provide an ideal natural laboratory for evaluating how rapid carbonate precipitation records isotopic disequilibrium across terrestrial environments and facies. Petrographic screening and mineralogical analyses were conducted on stratigraphically constrained samples to assess diagenetic overprinting. We interpreted preliminary $\delta^{13}\text{C}$, $\delta^{18}\text{O}$, and dual-clumped isotope (Δ_{47} – Δ_{48}) values within the context of depositional and diagenetic history to characterize departures from thermodynamic equilibrium. Preliminary petrographic characterization and initial dual-clumped results suggest variability among facies that may reflect formation mechanisms rather than temperature alone. These results will refine interpretations of carbonate-based paleotemperature reconstructions and establish a framework for identifying and quantifying isotopic disequilibrium in terrestrial systems during past greenhouse climates.

Investigating deep critical zone structure using landslide scars in the Taiwan Central Range

Lara Williams¹, Roman A DiBiase^{1,2}

¹Department of Geosciences, The Pennsylvania State University, University Park, PA, USA

²Earth and Environmental Systems Institute, The Pennsylvania State University, University Park, PA, USA

The Critical Zone (CZ) is the near-subsurface layer that integrates rock, water, air, and biotic processes; it is where bedrock is weathered and regolith is produced. The thickness of this weathered zone is dependent on lithology, climate, and erosion rate. Models indicate that weathering depths will be deeper in areas of slower erosion, as slower erosion rates will increase soil residence times and allow for deeper propagation of weathering. However, it is difficult to isolate the influence of individual factors, and the lower boundary of the CZ is difficult to study as it is often beneath tens of meters of weathered material.

To address these challenges and isolate the role of erosion in determining weathering depth, I plan to examine landslide scars in the Central Range of Taiwan that provide a unique window into the subsurface. The steep, rapidly eroding Central Range has a consistent climate and lithology, with landslides cutting directly into small patches of low relief topography found along the spine of the range. These landslide scars often expose the full depth of the weathered zone and extend into unaltered bedrock below. I plan to measure the depth of weathering in these scars using aerial photography and high-quality drone imagery obtained in 2024 from Mt. Qilai from which I will build 3D models of the topography. By using the slope of the low relief surfaces as a proxy for erosion rate, this will provide a first-order sense of how weathering depth responds to erosion.

A Network Analysis of Administrative Calls and User Interdependencies in the Colorado River Basin

Tsunami Sharmba¹, Dr. Antonia Hadjimichael¹

¹Department of Geosciences, The Pennsylvania State University, University Park, PA, USA

The Colorado River sustains over 40 million people, 5 million acres of cropland, and a \$1.4 trillion economy across seven U.S. states and Mexico. The basin faces mounting stress from chronic overallocation, prolonged drought, and accelerating climate change. Initial water allocations were based on flow estimates that exceeded actual averages by at least 3 million acre-feet, a structural deficit that has deepened as warming temperatures reduce snowpack and accelerate evaporation. As supply and demand diverge, the basin is approaching a governance crisis that current frameworks are ill-equipped to address.

Under prior appropriation, senior water rights holders place administrative calls that curtail junior users during scarcity. These calls are the operational backbone of the system, but the patterns they produce regarding who curtails whom, how often, for how long, and under what hydrological conditions, reveal how climate stress propagates unevenly across a rigid, seniority-bound network of interdependent users. When one user's call restricts another, that stress can trigger broader adaptive responses such as water transactions, crop changes, and decisions that carry their own risks and tradeoffs. Understanding where those pressures concentrate and how they cascade is essential for anticipating maladaptation before it locks in.

This research maps the structure of water user interdependencies in the Colorado River Basin using administrative call records from the Colorado Division of Water Resources. By representing users as a network, this work identifies which users and reaches are most vulnerable to cascading curtailment and where the system is most delicate. These findings will inform future efforts to build resilience into governance frameworks amid a deeply uncertain and changing climate.

Low-Cost Monitoring of Inert Gas Stimulation with DAS

Jackson Saftner¹, Xinxin He², Tieyuan Zhu¹, Shimin Liu²

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

²*Department of Energy and Mineral Engineering, The Pennsylvania State University, University Park, PA, USA*

High-density microseismic monitoring is a proven tool for stimulation verification and reservoir surveillance, but conventional campaigns typically require a dedicated monitoring well or a large surface geophone array. Here, we evaluate distributed acoustic sensing (DAS) as a cost-effective alternative by analyzing a continuous dataset acquired during an inert-gas (liquid CO₂) stimulation at an onshore well site near Olean, New York.

The dataset was recorded on a fiber array attached to the same wireline used to deploy the stimulation tool to the target interval. This single-well setup avoids an additional monitoring well and still provides dense, high-sensitivity measurements. In the continuous record, we capture the stimulation pulse with high signal-to-noise ratio and a rich post-stimulation interval characterized by sustained changes in seismic energy and numerous broadband impulsive signals consistent with cracking or deformation episodes.

To enable rapid diagnostics, we develop a lightweight detection workflow based on multi-band energy attributes and STA/LTA triggering. Applying the workflow, we identify 732 candidate impulsive events and track the post-stimulation event-rate evolution, finding an Omori-type decay similar to earthquake aftershocks. We then add a simple source-location step to estimate first-pass locations for a subset of detections (58 events), providing stimulation-verification metrics that indicate fracture propagation to approximately 50 m from the stimulation well.

Distributed Sensing of Frost Quakes and Ice-Wedge Cracking in Utqiagvik, Alaska

Gabriel Rocha dos Santos¹, Tieyuan Zhu¹

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

Ice wedge polygons are distinctive periglacial landforms that result from repeated cycles of ground contraction and crack infilling in permafrost terrains. However, the fracture events that initiate their growth remain poorly documented through direct field observations. In this study, we analyze thousands of short-duration seismic events recorded over two winters using continuous Distributed Acoustic Sensing (DAS) in Utqiagvik, Alaska. These events display clear body wave arrivals followed by dominant surface wave energy and broadband high-frequency content, suggesting shallow sources near the fiber. We found a strong correspondence between peaks in seismic activity and peaks in modeled ground thermal stress, derived from a thermo-viscoelastic model constrained by Distributed Temperature Sensing (DTS) from a co-located fiber. These signals are interpreted as cryoseisms resulting from the thermal cracking of ice wedges during periods of intense ground cooling, commonly referred to as frost quakes. To investigate the spatial variability of these events, we apply Matched Field Processing (MFP) using both surface and body wave arrivals. The resulting maps reveal evolving fracture patterns across the tundra surface. In addition, we performed source mechanism inversion on a subset of near-field events to investigate whether their waveforms can be used to infer rupture orientation or mode. Our results demonstrate the utility of DAS and DTS for detecting and characterizing thermally induced fracturing in permafrost. By combining seismic and thermal analyses, we provide new insights into the seasonal evolution of subsurface cracking and ice wedge development in a rapidly changing Arctic climate.

Listening to Arctic Sea Ice: Seismic Tremors as a Proxy for Ice Strength

Dayang Zhou¹, Gabriel Rocha dos Santos¹, Jackson Saftner¹, Tieyuan Zhu¹

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

Sea ice is a critical component of the polar system, supporting Arctic ecosystems and coastal communities, reflecting a substantial fraction of incoming solar radiation, and attenuating ocean wave energy by acting as a mechanical buffer along coastlines. Under ongoing climate warming, Arctic sea-ice extent has declined rapidly. Reduced ice cover increases coastal exposure to wave forcing, accelerating erosion and heightening the vulnerability of infrastructure and communities. Monitoring nearshore sea-ice stability is therefore essential for assessing coastal hazards and environmental change in the Arctic.

In contrast to remote sensing methods, which typically provide spatial resolutions of tens of kilometers and temporal resolutions of days to weeks, seismic observations offer continuous, high-temporal-resolution monitoring that operates reliably through polar night and severe weather. Seismic data are particularly sensitive to mechanical ice processes that are difficult to observe directly. Interactions between drifting pack ice and landfast ice generate distinctive harmonic tremor signals that can be separated from other seismic sources. In this study, we investigate the physical processes responsible for these harmonic features and explore their relationship to landfast-ice extent through analysis of fundamental frequencies.

To enable systematic detection, we apply a machine learning approach based on YOLO (You Only Look Once) to identify harmonic signals in the time–frequency domain. The detected events exhibit clear seasonal variability, suggesting strong links to the evolving state of nearshore sea ice. These results demonstrate the potential of seismic monitoring combined with machine learning to provide new constraints on coastal sea-ice dynamics.

Assessing Post-Collapse Flank Instability at Anak Krakatau Using Satellite Remote Sensing

Young Cheol Kim¹, Christelle Wauthier^{1,2}, Thomas R. Walter³, Semeidi Husrin⁴, Herlan Darmawan⁵, Daniel Müller³

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

²*Institute for Computational and Data Science, The Pennsylvania State University, University Park, PA, USA*

³*GFZ German Research Centre for Geosciences, Helmholtz Centre Potsdam, Telegrafenberg Potsdam, Germany*

⁴*The National Research and Innovation Agency (BRIN), Indonesia, Jakarta, Indonesia*

⁵*Laboratory of Geophysics, Department of Physics, Faculty of Mathematics and Natural Sciences, Universitas Gadjah Mada, Yogyakarta, Indonesia*

Anak Krakatau Volcano, Indonesia, experienced a major flank collapse on December 22, 2018, generating a tsunami that caused significant casualties. Prior InSAR studies showed more than a decade of instability on the southwest flank. This study investigates whether instability continued after the 2018 event. We used ALOS-2 and Sentinel-1 data spanning 2019 to 2024. Due to ongoing eruptive activity and surface changes, three separate time-series were generated to ensure enough reliable pixels for InSAR time-series analysis. The key focus area was the southwest part of Anak Krakatau, where faster movement was observed prior to the collapse and most of the newly erupted material had accumulated.

Post-collapse displacements show a strong gradient that aligns with fractures observed in drone imagery. To better interpret motion, we decomposed the InSAR time series into vertical and east–west components, excluding the north–south direction due to low sensitivity. The vertical displacement shows an exponential decay pattern, interpreted as compaction and consolidation of fresh deposits. To avoid this signal dominating source modeling, we limited geodetic modeling to later intervals of the time series. These displacements were then used in rectangular dislocation inversions to characterize the subsurface processes controlling post-collapse deformation.

Friday 17th April

Oral session 4

11:00 am – 12:00 pm

11:00 am – 11:15 am

Sierra Melton

Ph.D. student, 5th year, Post-Comps

Enhanced Grounding Zone Stresses at an Ice Cliff Compared to at a Freely Spreading Ice Shelf

11:15 am to 11:30 am

Noshin Sharmili

Ph.D. student, 4th year, Post-Comps

Exploring the Depositional & Erosional Morphology of Jezero Crater

11:30 am – 11:45 am

Caleb Norville

Ph.D. student, 4th year, Post-Comps

Impacts of Fire and Solar Radiation on Afroalpine Vegetation in the Rwenzori Mountains

11:45 am – 12:00 pm

Eric Hasegawa

Ph.D. student, 5th year, Post-Comps

The effect of ring substitutions on aromatic carboxyl oxygen isotope geochemistry and applications to macromolecular systems

Enhanced Grounding Zone Stresses at an Ice Cliff Compared to at a Freely Spreading Ice Shelf

Sierra Melton¹, Byron Parizek^{1,2}, Richard Alley¹, Sridhar Anandakrishnan¹

¹Department of Geosciences, and Earth and Environmental Systems Institute, Pennsylvania State University, University Park, PA, USA

²Mathematics and Geoscience, Pennsylvania State University, DuBois, PA, USA

Ice stored in the Greenland and Antarctic Ice Sheets largely flows to the ocean through fast-flowing outlet glaciers and ice streams, forming floating ice shelves when it crosses the grounding zone. Ice shelves are susceptible to melting, rifting, and collapse, which has been observed to speed flow of non-floating ice and to expose grounded ice cliffs that can retreat rapidly by iceberg calving, raising sea level. Most ice sheet models do not simulate coupled ice-shelf loss and calving-cliff retreat because of difficulty in modeling the governing processes at suitable spatiotemporal resolutions and parameterizing the sub-gridscale physics involved in these transition regions. With the Ice-sheet and Sea-level System Model (ISSM), we compare the 3D grounding zone stresses for a freely spreading ice shelf with those at an ice cliff for an idealized outlet glacier. At an ice cliff, lack of support from any further glacial ice in the down-flow direction causes a stress imbalance, compensated for by enhanced englacial stresses. These enhanced stresses should speed rates of glacial thinning and retreat for an ice cliff compared to a freely spreading shelf because they are occurring within the grounding zone rather than within the distal zones of the attached ice shelf. Even a freely spreading ice shelf underestimates cliff stresses by shifting stresses from the grounding line to the front of the shelf. Our results will ultimately help represent ice-cliff retreat more accurately in ice sheet models to improve sea-level rise projections.

Exploring the Depositional & Erosional Morphology of Jezero Crater

Noshin Sharmili¹, Liz Hajek¹, Kathryn Stack Morgan², Ben Cardenas¹, Libby Ives³

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

²*NASA Jet Propulsion Laboratory, Pasadena, CA*

³*Michigan Geological Survey, Kalamazoo, MI*

The Jezero crater western fan, an exploration site of the Mars 2020 Perseverance rover, contains a record of paleoenvironmental conditions, sedimentary processes, and potential biosignatures. It has generally been assumed that the modern shape and extent of the Jezero western fan is the result of the interplay between a deltaic deposit and billions of years of wind erosion on the Martian surface. Fine grained facies such as silt and clay would have been preferentially removed by wind erosion while rocks made of sand and gravel would have been more likely to be preserved.

However, this hypothesis is not consistent with the modern shape and extent of the Jezero western fan. Specifically, the scarp that defines the current extent of the western fan is abrupt and does not coincide with a downstream decrease in grain size. Instead, observations made during Perseverance's traverse across the crater floor and the Jezero crater western fan suggest that the strongest control on the present shape of the fan is the spatial distribution of a sandstone- and boulder-bearing sequence ("the Upper Fan group") that is unconformably overlain on underlying fluvial and deltaic sediments and absent on crater floor. Boulders are impervious to wind erosion and would essentially armor any underlying sedimentary rocks from erosion. This study employed numerical models to investigate the factors contributing to the preservation of Jezero Crater's western fan morphology and to understand the erosion rates affecting its formation and evolution. The experiments tested the hypothesis that the fan's shape was originally inherited from the lower Otis Peak Formation and later preserved by the erosion-resistant igneous boulder deposits of the upper Otis Peak Formation.

Impacts of Fire and Solar Radiation on Afroalpine Vegetation in the Rwenzori Mountains

Caleb Norville¹, Sarah Ivory^{1,2}

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

²*EESI, The Pennsylvania State University, University Park, PA, USA*

Previous work has elucidated the responses of Rwenzori Mountains Afroalpine vegetation to Holocene temperature and precipitation changes, suggesting that Afroalpine ecologies may be more insulated from the impacts of increasing temperature than lower Afromontane forest communities. Precipitation changes over the Holocene seem to correlate more closely to observed changes in vegetation recorded by pollen preserved in Afroalpine lakes. However, a large portion of the variance in the pollen data remains unexplained by these climate factors, necessitating the examination of other environmental or ecological factors as potential drivers of vegetation. Fire has been proposed as a major catalyst for rapid vegetation change in the Afroalpine zone, especially following a large wildfire in 2012. However, recent work has suggested that large Afroalpine fires were rare to absent across most of the Holocene, making the 2012 event unprecedented over the past 13 ka. Ongoing work is focused on understanding the record of vegetation response to this localized recent fire event.

Afroalpine elevations produce harsh conditions for plants, where frequent cloud cover and frost limits plant growth, and high UV-B influxes necessitate adaptations to this damaging solar radiation. Plants actively respond to changing UV-B influxes during the growing season by regulating their production of protective UV-B absorbing compounds (UACs) which have been shown to be well preserved within fossil pollen grains. These UACs in selected pollen taxa can be quantified using micro-FTIR spectroscopy, providing a pollen-based proxy for paleo UV-B influx. Changes in fossil pollen UACs can track orbitally controlled changes in solar insolation; however, the impacts of UV-B blocking cloud cover have yet to be examined. Ongoing work is focused on determining whether changes in cloud cover associated with the termination of the African Humid Period can be observed in fossil Afroalpine pollen, potentially providing a novel proxy for paleo cloud cover.

The effect of ring substitutions on aromatic carboxyl oxygen isotope geochemistry and applications to macromolecular systems

Eric Hasegawa¹, Jason Boettger², Katherine Dawson³, Katherine Freeman¹, Max Lloyd¹

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

²*North Carolina Department of Environmental Quality, Raleigh, NC, USA*

³*Department of Environmental Sciences, Rutgers University, Newark, NJ, USA*

The position-specific oxygen isotope geochemistry of macromolecules such as lignin, kerogen, or meteoritic insoluble organic matter (IOM) can give insight into the origins and reaction pathways of different functional groups. These functional groups are dominantly associated with aromatic moieties in the macromolecular structures. However, it is unclear if and how functional group electronic properties differ due to substitutions on aromatic rings and how they result in different isotope geochemistry. An example is the oxygen exchange chemistry that can occur on an aromatic carboxyl group such as the one found in benzoic acid. In this case, the carbon is sp^2 hybridized, which links the oxygen to the larger aromatic framework. We aim to answer the question: how do additional ring substitutions change the oxygen isotopic exchange chemistry at the carboxyl site, and how does this affect our interpretation of carboxyl group chemistry in natural samples? Initial density functional theory modeling suggests little to no difference in fractionation factors, and we follow these studies with physical equilibration experiments using methods developed in our previous study of phenol-water oxygen exchange. We reacted benzoic acid and vanillic acid with waters of known oxygen isotopic compositions at elevated temperatures, and we injected these solutions into our electrospray ionization-Orbitrap mass spectrometer to determine oxygen equilibrium fractionation factors (the difference between an equilibrated carboxyl group's isotopic composition and the water's isotopic composition) for these two molecules. Carboxyl groups can exchange their oxygen in low pH conditions, and therefore their isotopic composition, once corrected with a fractionation factor, will likely represent the isotopic composition of the last acidic alteration fluid to which they were exposed. The impact of additional aromatic functional groups will be used as an analog to better understand the impact of aromatic linkages in macromolecular frameworks.

Friday 17th April

Oral session 5

1:00 pm – 2:15 pm

1:00pm – 1:15 pm

Erik Schoonover

Ph.D. student, 5th year, Post-Comps

Linking Arc Granitoids to Ore Deposits through Fluid–Melt–Rock Interactions

1:15 pm to 1:30 pm

Emma Hartke

Ph.D. student, 4th year, Post-Comps

Holocene climate changes, ecosystem conditions & early human activity in the Central Mediterranean: a multi-proxy investigation

1:30 pm – 1:45 pm

Juliana Drozd

Ph.D. student, 4th year, Post-Comps

Lipid Carbon Isotope Patterns as Metabolic and Environmental Proxies

1:45 pm – 2:00 pm

Em White

Ph.D. student, 4th year, Post-Comps

Reworking of Hadean crust: Evidence from $\mu^{142}\text{Nd}$ anomalies in later Archean crust

2:00 pm – 2:15 pm

Rory Changleng

Ph.D. student, 4th year, Post-Comps

Disentangling Polygenetic Zircon Populations from Archean Gneisses Reveals a New Eoarchaeon Terrane

Linking Arc Granitoids to Ore Deposits through Fluid–Melt–Rock Interactions

Erik Schoonover¹, Cole J. Stern¹, Renan Beckman¹, Rory Changleng¹, Evan Ritchey¹

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

Rising demand for key metals has placed increased focus on discovering new deposits in Earth's crust. Resources, such as Fe, Cu, and the rare earth elements, are fundamental for infrastructure, energy systems, and national security. Generating deposits large enough to be economically viable requires efficient crustal processes that mobilize and concentrate these metals of interest. Investigating the mechanisms of ore-forming processes is important, particularly, for developing new exploration targets and reducing the economic costs associated with ore discovery and characterization. Crustal ore deposits provide a partial view of these processes because they are the final product of multistage, complex processes. Deposits are often overprinted by successive mineralization events, strongly modified by post-emplacement alteration, and their source rocks are rarely preserved or exposed. As a result, the physical and chemical mechanisms of ore formation are difficult to observe directly.

Bridging this gap in deposit formation requires shifting focus from the mineralized deposits to the magmatic environments that generate them. Two major deposit types that supply strategic and critical elements are iron oxide-apatite (IOA) and iron oxide-copper-gold (IOCG) deposits. These deposits are exemplar models for metal concentration and mobilization from arc magmas. Competing models for these deposits suggest either that enrichment occurs through separation of an Fe-rich immiscible melt or through Cl-rich fluid flux that sequester metals from a crystal-rich magma. Therefore, without source context, the interactions and relative roles of fluid, melt, and magmatic rocks during metal concentration remains unresolved.

My current hypothesis is that metal-enriched, channel-like features in arc granitoids represent the preserved record of metal mobilization during the magmatic-hydrothermal transition. These features could provide a means to directly link late-stage arc magmatism with ore-forming processes. The Tuolumne Intrusive Suite (TIS), CA, USA is an ideal setting, enabling clear observation of the parameters governing fluid-melt-rock interaction during the metal mobilization.

Holocene climate changes, ecosystem conditions & early human activity in the Central Mediterranean: a multi-proxy investigation

Emma Hartke¹, Kate Freeman¹, Miquela Ingalls¹, Melissa Berke²

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

²*Civil and Environmental Engineering and Earth Sciences, Notre Dame University, Notre Dame, IN, USA*

Holocene records of environmental change capture the interrelated evolution of climate, ecosystems, and early human activity, especially as societies in the Central Mediterranean transitioned toward increasingly settled agropastoral lifestyles during the Middle Holocene. Disentangling the relative influences of postglacial climate variability and anthropogenic landscape modification remains challenging, particularly in this climatically sensitive region. Continuous sedimentary records from both terrestrial and marine archives were analyzed using multiple biogeochemical and climate proxies to reconstruct these parameters and relationships in the Central Mediterranean.

This work integrates organic (lipid biomarker, compound-specific isotope) and carbonate clumped isotope analyses from terrestrial (coastal Croatian lake) and marine (Aegean Sea) sediment cores to assess changes in vegetation dynamics, fire activity, human lifestyles, temperature, and hydroclimate across the Central Mediterranean. Together, these records capture coupled ecological, climatic, and human-driven changes across the Holocene (11,700 kyrs–present), a period for which regional paleoclimate data remain sparse.

Terrestrial biomarker results (*n*-alkanes, PAHs, fecal stanols, GDGTs) spanning the Middle Holocene indicate intensifying human activity coincident with wetter regional conditions, the combination of which led to a temporary, localized enhancement of microbial activity. To contextualize these results within the regional climate space, we applied proxies for temperature (GDGTs, carbonate clumped isotopes) and hydrology (compound-specific hydrogen isotopes) to both our terrestrial and marine records. Results indicate a warming, drying trend throughout the Holocene, with the Middle Holocene a ‘transitional’ period tentatively punctuated by alternating wet/dry cycles. Ongoing hydrogen isotope analyses will better constrain the role of changing hydroclimate conditions on landscape evolution and human activity in the Central Mediterranean.

By pairing biogeochemical indicators of climate variability, ecosystem conditions, and early human activity, this study 1) provides a pathway for tracing the spatiotemporal evolution of human lifestyle modes during a period of transitional climate dynamics, and 2) highlights the sensitivity of Central Mediterranean landscapes to even modest climatic perturbations and anthropogenic impacts, offering a long-term perspective essential for assessing future environmental change in this rapidly warming, drying region.

Lipid Carbon Isotope Patterns as Metabolic and Environmental Proxies

Juliana Drozd¹, Katherine Freeman¹, Elliott Mueller²

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

²*Department of Geological Sciences, University of Colorado Boulder, Boulder, CO, USA*

The chemical structures and stable isotope compositions of lipids preserved in the rock record, known as biomarkers (molecular fossils), serve as proxies for properties that can be used to reconstruct past ecology and climates. Reconstructions based on biomarker proxies rely on knowledge of how organisms, specifically primary producers, change their lipid production in response to growth conditions. Metabolic regulation triggered by environmental change is difficult to assess with the biomarker record. However, differences in the carbon isotope compositions between different types of lipids potentially capture signs of metabolic regulation because the flux of metabolites through branch points in biosynthetic pathways can control the expression of isotope effects by enzymes. Fluxes are regulated in response to an organism's needs and its environmental conditions but are also subject to constraints from the organism's physiology, including the compartmentalization of eukaryotic cells. We used metabolic reaction network isotopologue modeling and comparison to existing datasets to assess these signatures for green algae, haptophyte algae, cryptomonad algae, and diatoms. In many algae, fatty acid biosynthesis is upregulated due to environmental stresses like temperature changes and nutrient starvation. This results in a decreased difference in the carbon isotope compositions of fatty acids and phytol, an important isoprenoid lipid. For green algae, sterol biosynthesis is co-regulated with fatty acid biosynthesis, which results in sterols becoming ¹³C-enriched relative to phytol when the algae are under environmental stress. We can potentially use the differences in the $\delta^{13}\text{C}$ values of different types of lipids as a proxy for environmental stress-induced metabolic regulation in eukaryotic algae. These findings for modern algal cultures might be able to be expanded to the biomarker record to spot paleoenvironmental conditions that would have stressed ancient algal populations.

Reworking of Hadean crust: Evidence from $\mu^{142}\text{Nd}$ anomalies in later Archean crust

Emily L. White¹, Michelle Jordan², Steven B. Shirey², Timothy Mock², D. Graham Pearson³, William J. Davis³, Jesse R. Reimink¹

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

²*Earth and Planets Laboratory, Carnegie Institution for Science, Washington, DC, USA*

³*Department of Earth and Atmospheric Sciences, University of Alberta, Edmonton, AB, Canada*

⁴*Natural Resources of Canada, Geological Survey of Canada, Ottawa, ON, Canada*

Earth's dynamic lithosphere sets it apart from other observable silicate planetary bodies, as it features active and mobile-lid plate tectonics. However, the evolution/onset of these modern-styled plate tectonics, which have shaped Earth into the habitable world it is today, remains a fundamental and widely debated topic in the Earth sciences. The rate of net-continental growth throughout Earth history is required to evaluate the evolution of these geodynamical processes. One consequence of tectonics is how over billions of years of Earth history the continental crust is often altered, overprinted, and/or reworked. This has led to a paucity of ancient crustal material preserved at the surface, limiting our record of Earth's earliest continental crust and making it difficult to evaluate crustal growth through time. However, developments in high-precision whole-rock analysis of the short-lived ^{146}Sm - ^{142}Nd isotopic system allow us to trace anomalous signatures of ancient precursors in the products of later magmatism, enhancing this record.

The Slave Craton (Canadian Shield) hosts excellent candidates for observing ancient crustal reworking, due to its near-constant record of episodic Archean crustal growth. It features 4020-2850 Ma gneisses in the Central Slave Basement Complex (including the oldest known zircon-bearing rock in its Acasta Gneiss Complex) and in the Tree River region. Additionally, 2650-2580 Ma granitoids occur across the province, with previous Pb-isotope and long-lived Nd-isotope data indicating derivation from both ancient and juvenile sources. We present $\mu^{142}\text{Nd}$ ($^{142}\text{Nd}/^{144}\text{Nd}$ normalized to the bulk terrestrial composition) values of fourteen Neoarchean and Meso- to Eo-Archean gneisses from across the Slave Craton. Our results document a more widespread presence of Hadean signatures in Eoarchean crust than previously reported (ca. -16 ppm $\mu^{142}\text{Nd}$), and some anomalous Neoarchean granitoids suggesting possible derivation from ancient crustal reworking. These results will test for widespread reworking of Hadean continental crust, improving our understanding crustal growth through time.

Disentangling Polygenetic Zircon Populations from Archaean Gneisses Reveals a New Eoarchaeon Terrane

Rory Changleng¹, Erik J. Schoonover¹, Cristy Stoian^{1,2}, Joshua M. Garber^{1,3}, D. Graham Pearson⁴, Yan Luo^{4,5}, Jesse R. Reimink¹

¹*Department of Geosciences, Pennsylvania State University, PA, USA*

²*Department of Geology, Colorado School of Mines, CO, USA*

³*School of Earth and Environmental Sciences, University of St Andrews, St Andrews, UK*

⁴*Department of Earth and Atmospheric Sciences, University of Alberta, Edmonton, AB, Canada*

⁵*Ken Williams Radiogenic Isotope Geoscience Laboratory, Texas A&M University, College Station, TX, USA*

A principal limitation of our understanding of the early Earth is the paucity of samples preserved from this time. Only around a dozen locations worldwide retain crust from the Eoarchaeon (>3.6 Ga), and the true ages of many of these ancient vestiges of our planet remain hotly debated. One of the best recorders of crustal formation and alteration events is the mineral zircon. However, distinguishing between multiple igneous and metamorphic events experienced by a zircon population within an Archaean gneiss defines a critical question: how old is this rock?

Each tract of ancient crust discovered carries a disproportionate significance in shedding light on the magmatic and geodynamic conditions shaping our planet's earliest evolution. Consequently, careful evaluation is required when assessing whether a terrane preserves such ancient lithologies. We present a new U-Pb zircon geochronology from three basement gneisses from the Eokuk Uplift, northwest Slave Craton. This region represents an excellent target to search for new remnants of old crust due to superb outcrop exposure, proximity to the 4.03 Ga Acasta Gneiss Complex, and its relatively understudied nature resulting from its remote Arctic setting.

All three gneisses are complex and polydeformed, with zircon age populations spanning more than 500 million years. We employ depth-profile analysis to target zircon rim domains and integrate these data with cathodoluminescence SEM imaging to aid in interpreting conventional LA-ICP-MS spot analyses. This combined approach allows us to reconstruct zircon crystallisation and alteration histories for each sample.

Although Eoarchaeon-aged zircons occur in all three gneisses, only one sample preserves a primary igneous zircon population recording the original crystallisation age of the host rock. At 3.81 Ga, this sample demonstrates that Eoarchaeon crust is more widespread within the Slave Craton than previously recognised, significantly expanding the material available for studying the earliest chapters of Earth's history.

Friday 17th April

Oral session 6

2:45 pm – 4:00 pm

2:45 pm – 3:00 pm

Fran Meyer

Ph.D. student, 4th year, Post-Comps

Oxidized uranium in organic-rich shales from reducing environments

3:00 pm – 3:15 pm

Hanna Leapaltd

Ph.D. student, 4th year, Post-Comps

Signatures of microbial carbon cycling in magnesium carbonate minerals

3:15 pm – 3:30 pm

Eddie Spagnuolo

Ph.D. student, 4th year, Post-Comps

From aquatic ferns to rainforest trees: The first plant macrofossils from the early Eocene Ghazij Formation record a diverse flora and associated insect herbivory on the Indian Subcontinent

3:30 pm – 3:45 pm

Dani Buchheister

Ph.D. student, 4th year, Post-Comps

The iron-sulfur world in a bottle: FeS₂ formation in microbial enrichment cultures

3:45 pm – 4:00 pm

Sarah Jonathan

Ph.D. student, 5th year, Post-Comps

Measuring weathering rates across elevational gradients in climate and slope in the Sierra Nevada and Inyo Mountains of California

Oxidized uranium in organic-rich shales from reducing environments

Fran Meyer¹, Anthony Chappaz², Dan Gregory³, Eve Carrothers³, Erik Sperling⁴, Justin Strauss⁵,
Lucien Nana Yobo⁶, Kimberly Lau¹

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

²*Department of Earth and Atmospheric Sciences, Central Michigan University, Mount Pleasant, MI, USA*

³*Department of Earth Sciences, University of Toronto, Toronto, ON, CA*

⁴*Department of Earth and Planetary Sciences, Stanford University, Stanford, CA, USA*

⁵*Department of Earth Sciences, Dartmouth College, Hanover, NH, USA*

⁶*Department of Geology and Geophysics, Texas A&M University, College Station, TX, USA*

The availability of oxygen in the Earth's ancient ocean and atmosphere is a key parameter for understanding Precambrian and Paleozoic biotic evolution and nutrient cycling. Due to the lack of direct oxygen measurements in deep time, we must turn to the geochemical signatures preserved in the sedimentary record. The enrichments and isotope ratios of redox-sensitive elements deposited in marine basins, like U, provide a way to reconstruct changes in ocean oxygen availability. However, these proxies rely on many unverified assumptions of how their solubility, speciation, and burial are controlled by redox conditions. Currently, interpreting U in reducing marine shales as a paleo-redox proxy assumes that burial and isotope fractionation are only associated with the reduction of U(VI) to U(IV), and sequestration of U(VI) and U(V) is negligible. Yet, the burial pathways and speciation of U in reducing marine environments still need to be defined. Without characterizing the oxidation states or phase associations of U in marine shales, interpretations of the U-based proxies remain uncertain.

In this study, we use synchrotron-based X-ray fluorescence imaging and spatially resolved X-ray absorption near edge spectroscopy (XANES) at the U L3-edge paired with high energy resolution fluorescence detection XANES at the U M4-edge to determine the spatial distribution and speciation of U in marine shales from multiple localities and time intervals. Samples included outcrop and drill core shales from the Cretaceous Eagle Ford Group, the Devonian Marcellus Formation, and the Ordovician Road River Group. XANES spectra at both the U L3- and M4-edge suggest that three U oxidation states (IV, V and VI) are present, despite proxy data suggesting anoxic depositional settings. The ubiquitous presence of U(V) and U(VI) in marine shales from reducing environments adds new understanding of the U marine cycle and challenges the current use of U as a paleo-redox proxy.

Signatures of microbial carbon cycling in magnesium carbonate minerals

Hanna C. Leapaltd¹, Tori C. Cassady², Frank A. Corsetti², Miquela Ingalls¹

¹*Department of Geosciences, The Pennsylvania State University, Pennsylvania, 16802, USA*

²*Department of Earth Sciences, University of Southern California, California, 90089, USA*

Magnesium carbonate minerals have been identified in the Jezero Crater paleolake basin on Mars and have been collected by *Perseverance* to be evaluated for potential biosignatures. However, there is not yet a diagnostic set of textural, geochemical, and isotopic characteristics that can distinguish biotic from abiotic magnesium carbonate minerals. We compared magnesium carbonate samples that form biotically – from microbial mats and shallow sediments in two hyperalkaline lake/playa systems in British Columbia (South Atlin Playa and Alberta Lake) – to those forming abiotically from carbonation of serpentinite in bedrock. We used a novel isotope system (dual-clumped isotopes; Δ_{47} - Δ_{48}) and hypothesized that biological carbon cycling within the microbial mats will cause kinetic isotope effects resulting in Δ_{47} - Δ_{48} disequilibrium signatures. We measured the Δ_{47} and Δ_{48} values every 5 cm of 15 cm cores through the microbial mats as well as from the subsampled bedrock. All microbial mat cores revealed the same, distinctive trend with depth: the Δ_{47} - Δ_{48} values of samples in the upper 5 cm of the cores within microbial mat, where we expect active cyanobacterial photosynthesis, yielded the largest offset from the Δ_{47} - Δ_{48} theoretical equilibrium. The microbial mat sample Δ_{47} - Δ_{48} values yield disequilibrium signatures expected for association with photoautotrophy. We interpret the observed values to be a signature of cyanobacterial photosynthesis at the mat surface causing CO₂ degassing, leading to the observed $-\Delta_{47}/+\Delta_{48}$ kinetic isotope effects. Fluorescently labelled embedded core images from the upper 5 cm at both locations reveal cellular material incorporated into the fabric of the sediment, providing evidence of a microbial control on carbon cycling. If such kinetic isotope effects are absent from abiotically formed magnesium carbonate minerals, dual-clumped isotopes can provide a tool to help identify biogenicity of magnesium carbonate minerals when paired with other evidence like textural analysis.

From aquatic ferns to rainforest trees: The first plant macrofossils from the early Eocene Ghazij Formation record a diverse flora and associated insect herbivory on the Indian Subcontinent

Edward J. Spagnuolo¹, Peter Wilf¹, P. J. Przybylski¹, L. Alejandro Giraldo¹, Munir ul-Haq², Scott L. Wing³, William C. Clyde⁴

¹*Department of Geosciences, Pennsylvania State University, University Park, Pennsylvania, 16802, USA.*

²*Geological Survey of Pakistan, Islamabad, Pakistan.*

³*Department of Paleobiology, National Museum of Natural History, Smithsonian Institution, Washington DC, 20560, USA.*

⁴*Department of Earth Sciences, University of New Hampshire, Durham, New Hampshire, 03824, United States.*

Well-dated plant fossils from tropical Asia are rare, limiting understanding of how and when this region attained its floral composition and high species richness. Here, we report the first plant macrofossils and associated insect damage from the early Eocene Ghazij Formation (Balochistan, Pakistan), well known for its mammalian faunas that document early biotic interchange associated with the India (Indo-Pakistan)-Asia collision. The macrofossils were collected from 15 sites from the middle and upper Ghazij, totaling 420 compression specimens, primarily of leaves, dating to the early Eocene Climatic Optimum (EECO). The flora has an estimated 40 species, dominated by Sapindaceae (maple, lychee family) leaflets affiliated to the rainforest tree *Pometia*, and multiple Fabaceae (legume family) and Arecaceae (palm) morphotypes. The *Pometia*-like leaflets are the oldest macrofossils potentially affiliated to the genus and would support its proposed Indian biogeographic history based on prior pollen data. Rarer morphotypes include Menispermaceae (moonseeds), possible Rhamnaceae (buckthorns), possible *Calophyllum*, Araceae (*Monstera* family), and the aquatic ferns *Acrostichum* and *Salvinia*. Insect-feeding damage is diverse and abundant, totaling 47 damage types (DTs), including nine hole feeding DTs, seven piercing and sucking DTs, 13 gall DTs, and six mines. Adjusted for sample size, the Ghazij flora has some of the highest herbivory levels in the fossil record (although mining is rare), providing novel information about tropical plant-animal interactions during the EECO.

The iron-sulfur world in a bottle: FeS₂ formation in microbial enrichment cultures

Dani Buchheister¹, Kenneth Broad², Nadir Quarta³, Julie Cosmidis⁴, Jennifer Macalady¹

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

²*Department of Environmental Science and Policy, University of Miami, Coral Gables, FL, USA*

³*QUARTA Dimensione, Switzerland*

⁴*Department of Earth Sciences, University of Oxford, Oxford, UK*

The Frasassi cave system in central Italy hosts a permanently stratified sulfidic aquifer with a shallow microoxic zone and a deeper anoxic zone. The deep aquifer lacks most of the oxidants used for respiration by surface life, and has sulfate concentrations of approximately one-eighth of marine, making it a promising analog for Proterozoic oceans. Biofilms in the deep aquifer contain abundant pyrite (FeS₂) mineral grains and a treasure trove of microbial “dark matter,” microorganisms with only distant relatives in public databases. There are multiple formation pathways for FeS₂, and the role that microbes play in iron sulfide mineral nucleation and crystal growth is the topic of ongoing debate.

In February 2023, Frasassi deep aquifer biofilms were retrieved to inoculate enrichment cultures targeting a “missing” metabolism—Wächtershäuser’s proposed pyrite- and hydrogen-producing metabolism that reacts iron sulfide (FeS) with hydrogen sulfide. After a year of incubation, clusters of cells and microspheroidal mineral grains were observed with transmitted light and epifluorescence microscopy. Further imaging and mineral analyses were performed using Raman spectroscopy and SEM-EDS. DNA extracted from the biofilm (inoculum) and the enrichment cultures were analyzed using 16S rRNA amplicon sequencing to determine the taxonomy of the major populations. Metagenomic sequences were also obtained for the enrichment culture. Raman and SEM-EDS data indicate the presence of FeS₂ in the enrichment culture bottles but not in the abiotic controls. FeS₂ mineral grains are microspheroidal but do not look like microbially-mediated pyrite produced in other laboratory studies. The enrichment communities show changes in taxonomic structure from the native biofilm and are much less diverse. Known sulfur cyclers are abundant in the bottles, yielding clues to the potential role of the enrichment microbes in FeS₂ formation.

Measuring weathering rates across elevational gradients in climate and slope in the Sierra Nevada and Inyo Mountains of California

Sarah R. Jonathan¹, Roman A. DiBiase¹, Lee B. Corbett², Paul Bierman²

¹*Department of Geosciences, The Pennsylvania State University, University Park, PA, USA*

²*University of Vermont, Rubenstein School of Natural resources and the Environment*

In mountain regions, where steep topography dominates, hillslopes often have a mixed surface cover, with patches of soil interspersed with bare bedrock. While this patchy surface cover is prevalent, the processes that control the breakdown – and thereby determine the persistence and extent of soil cover – are poorly constrained. Ultimately, the balance between soil loss, consisting of both physical erosion and chemical weathering, and soil production determines the extent of soil cover. However, both weathering and erosion can vary dramatically and may be influenced by a variety of factors, including climate, hillslope morphology, and tectonic setting. Here, I examine the extent and rate of weathering across ridgeline transects in three California sites spanning a range of bedrock exposures, slopes, and elevations: The Olancha Peak and Sacatar Wilderness areas in the Eastern Sierra Nevada and McElvoy Canyon in the Inyo Mountains. All three sites are underlain by granitic bedrock and experience similar altitudinal variations in climate and vegetation, but have different patterns of bedrock exposure. In the Sacatar Wilderness bedrock exposure is low and increases with elevation, while McElvoy Canyon demonstrates an inverse pattern of decreasing bedrock with increasing elevation. The Olancha Peak area occupies a middle ground with high bedrock exposure at all elevations. Across these landscapes, I use paired soil and bedrock samples to measure the extent of chemical weathering and, combined with a slope-based erosion rate model, calculate weathering rates. These values are combined with Lidar analyses of bedrock exposure and hillslope morphology and measures of soil depth to explore how weathering rates and soil production vary with soil cover across elevations. Initial results suggest that the extent of chemical weathering initially increases, then decreases with increasing elevation and the variability in weathering increases as slope steepen to the point of significant bedrock exposure.